

PERFORMATIVE PRIVACY: WHEN DIFFERENTIAL PRIVACY MAXIMIZES UTILITY

Uddalak Mukherjee

Dauphine PSL,

uddalakmukherjee49@gmail.com

Edwige Cyffers

CNRS Dauphine PSL,

edwige.cyffers@cnrs.fr

Yann Chevalereyre

Dauphine PSL,

yann.chevalereyre@lamsade.dauphine.fr

ABSTRACT

Privacy-preserving learning is often motivated by the idea that protecting users' data can preserve trust and thus participation, improving utility in the long term. However, this claim has not been formalized so far. In parallel, performative learning provides a framework for studying learning systems whose deployment affects the data they later observe. In this work, we bring these two perspectives together and introduce *performative privacy*, where data leakage reduces future participation. We study a simple model where agents repeatedly contribute data for mean estimation but may leave the system when their data is leaked. Privacy is implemented through differentially private mechanisms, creating a trade-off between estimation noise and future participation. We show, through a theoretical study of the dynamics and numerical experiments, that a finite privacy budget can outperform non-private estimation in the long term when the feedback loop between leakage and participation is sufficiently strong. This provides first evidence that differential privacy can be optimal not only as a protection mechanism, but also from the perspective of long-term utility.

1 INTRODUCTION

Protecting privacy in machine learning is motivated by moral, ethical, and practical reasons (Solove, 2008). Access to sensitive data, such as health records, is sometimes illegal without adequate protections (European Parliament & Council of the European Union). However, in many scenarios, privacy protection is motivated by users' willingness to participate in training, based on the risks it entails for their business or their own privacy if their data is leaked (Nissenbaum, 2004; IBM & Ponemon Institute, 2025). Users who generate training data can be seen as agents who decide whether to participate depending on the risk that their data is leaked. A natural feedback loop to consider is thus that, if a user has their data leaked, they may no longer be willing to participate in the future, harming future rounds of training. This is often claimed informally: privacy-preserving machine learning should optimize long-term utility by maintaining users' trust, and thus their data (Dwork & Roth, 2013). Surprisingly, this claim that private learning can provide optimal long-term utility has, to the best of our knowledge, never been formalized. We propose first results in this direction.

The gold standard for protecting data is differential privacy (DP) (Dwork et al., 2006). DP formalizes the following promise: a single data point should not influence the output of the algorithm too much, ensuring that any attacker with access to the output cannot gain significant information about the data point by observing the output. This protection, however, can only be achieved through randomization, often implemented by injecting noise into the algorithm. The level of privacy protection is then summarized by the privacy budget ϵ : the smaller the privacy budget, the stronger the privacy guarantee, and the larger the noise required, which decreases the accuracy of the model, creating a privacy-utility trade-off. The right value for ϵ is often debated in the community, based on ad hoc perceptions of data sensitivity and risk (Ponomareva et al., 2023; Cyffers, 2025; Lee & Clifton,

2011). Here, we take an orthogonal perspective, considering the optimal privacy budget to be the one that maximizes long-term utility.

Taking into account how models change the data over time is the motivation of *performative learning* (Perdomo et al., 2020), where the data distribution is parametrized by the same parameter as the model, and the goal is to optimize the overall system rather than find the best model for a fixed distribution, as usually done in supervised learning. Performative learning is an increasingly popular framework, motivated by examples where users may game the system by modifying their features to obtain more favorable outcomes, by settings involving generative models where data can be contaminated over time, or by scenarios where sampling can reinforce existing biases, raising long-term fairness concerns (Hardt & Mendler-Dünnér, 2023). Feedback loops due to privacy leakage have not been studied so far, but they constitute a natural dynamic to study, as it is well known that data leakage can cause trust issues and reduce participation, or even lead participants to adopt malicious behaviors (IBM & Ponemon Institute, 2025). Relatedly, the term *performative privacy* was coined by legal scholars (Skinner-Thompson, 2017) to describe the actions agents may take to exercise their right to privacy, by making unavailable information that could otherwise be available by default.

In this work, we design and study a simple model that allows us to quantify regimes in which a finite privacy budget maximizes utility in a repeated mean estimation problem. Rather than using the privacy budget directly as a proxy for the decision to participate, we use whether the data point has been leaked in the feedback loop. This makes our setting applicable to non-differentially private mechanisms as well and avoids focusing on a meta-interpretation of the privacy budget, instead focusing on actual algorithm outputs. More precisely, our contributions are the following:

- We propose a new object of study, the long-term utility of repeated learning tasks involving privacy-sensitive data, which we call *performative privacy*.
- We first study a simple setting with a set of agents who each hold a private binary value at each round and may withdraw from the learning process if their correct private value is used by the system, and in which additional agents may join at each step.
- We extend this setting to the computation of a d -dimensional average over time, with a more realistic definition of data leakage based on membership inference attacks.
- We show that a finite privacy budget can be optimal in both scenarios, proving that privacy can maximize utility in the presence of a feedback loop in terms of participation.

2 RELATED WORK

While a direct link between data leakage and future participation has not yet been formalized, differential privacy originates from Warner (1965), where randomization aims to reduce bias in data collection, and thus improve data quality. Inspirations from sociology, notably through the contextual integrity framework, have also been numerous (Nissenbaum, 2004; Cyffers, 2025; Cummings & Durfee, 2020). Considering data producers as privacy-aware agents is a popular framework, sometimes formalized as a data market (Ghosh & Roth, 2011; Pai & Roth, 2013; Fallah et al., 2022), in particular for deciding participation in surveys (Roth & Schoenebeck, 2012; Ligett & Roth, 2012). The choice of the privacy budget itself has been analyzed through economic and mechanism-design lenses (Hsu et al., 2014; Nissim et al., 2012; Cummings et al., 2015; Ligett et al., 2017), possibly personalizing it across users (Jorgensen et al., 2015). However, privacy awareness is often translated into a supposedly known optimal privacy budget, computed from expected utility and privacy preferences (Cummings et al., 2022; Chaudhuri et al., 2025), rather than modeled as a dynamical system. Participation has also been studied as a strategic decision in collaborative and federated learning (Donahue & Kleinberg, 2021; Blum et al., 2021; Karimireddy et al., 2022), without the link to privacy leakage that we consider. On the empirical side, data breaches measurably affect customer behavior and firm value (IBM & Ponemon Institute, 2025), and the way privacy guarantees are communicated changes users’ willingness to share their data (Xiong et al., 2020; Cummings et al., 2021), supporting the feedback loop we model.

Performativity in Trustworthy Machine Learning. Performative learning (Perdomo et al., 2020) is motivated by the social impact of machine learning deployments (Hardt & Mendler-Dünnér, 2023; Hardt et al., 2022), with dedicated optimization algorithms (Mendler-Dünnér et al., 2020) and regret

analyses (Jagadeesan et al., 2022; Zhu et al., 2025), and with links to robustness (Cyffers et al., 2024) and fairness (Somerstep et al., 2024). Closest to our setting are retention dynamics, where user groups shrink after repeatedly receiving poor utility (Hashimoto et al., 2018; Zhang et al., 2019); we study the privacy analogue, where users leave after a leakage event. As the population size depends on the whole history of deployments, our model is an instance of stateful performativity (Brown et al., 2022) rather than of the memoryless distribution map of Perdomo et al. (2020). Self-reinforcing feedback loops have also been documented in predictive policing (Ensign et al., 2018), dataset bias amplification (Taori & Hashimoto, 2023), and generative model collapse (Shumailov et al., 2024). Interestingly, participants can also contribute to steering the model toward desirable properties, as studied in collective action (Hardt et al., 2023). To the best of our knowledge, there is no theoretical work, nor simulated feedback-loop model, tackling privacy concerns from this perspective so far.

Membership Inference Attacks. Membership inference attacks (MIAs) were introduced in the context of genomic studies by Homer et al. (2008), who demonstrated that aggregate statistics can reveal an individual’s participation in a dataset. MIAs have played an important role in motivating DP and establishing lower bounds for private data analysis (Bun et al., 2014; Nasr et al., 2021). In machine learning, numerous MIAs have been proposed under different threat models and access assumptions (Shokri et al., 2017; Sablayrolles et al., 2019; Even et al., 2026; Carlini et al., 2021; Yeom et al., 2017; Ye et al., 2022). These attacks are also widely used to empirically audit machine-learning systems and evaluate privacy defenses (Jagielski et al., 2020; Aerni et al., 2024; Jayaraman & Evans, 2019; Thudi et al., 2022). The membership inference game is naturally cast in the hypothesis-testing formulation of DP (Wasserman & Zhou, 2010; Kairouz et al., 2015), culminating in Gaussian DP (Dong et al., 2022), whose privacy-loss random variable is the leakage statistic we analyze in Section 4.4. Beyond measuring average privacy leakage, recent work has shown that privacy risk can vary substantially across individuals (Leemann et al., 2024; Azize & Basu, 2025), motivating per-instance guarantees and individual privacy accounting (Wang, 2019; Feldman & Zrnic, 2021; Yu et al., 2023; Koskela et al., 2023), and that worst-case privacy guarantees may be overly pessimistic in finite-sample regimes (Haghifam et al., 2025). Most closely related to our setting, recent work has studied membership inference under repeated interactions (Michel et al., 2026).

3 WARM-UP: ESTIMATION OF BERNOULLI PARAMETERS

In this section, we derive the behaviors arising in the estimation of a Bernoulli parameter when the ground truth changes at each time step. The Bernoulli parameter is estimated from independent and identically distributed samples that are locally privatized using randomized response (RR). An answer is considered leaked whenever the true answer is preserved by RR.

Learning task We consider a repeated mean estimation problem over T time steps. At each step t , the learner computes an estimate $\hat{\mu}_t$ of the true parameter μ_t of an underlying Bernoulli distribution. The learner has access to possibly privatized responses $(Y_{i,t})_{i=1}^{N_t}$ of the users’ samples $(X_{i,t})_{i=1}^{N_t} \in \{0, 1\}^{N_t}$, drawn i.i.d. from $\mathcal{B}(\mu_t)$. The learner aims to minimize the mean squared error of the parameter estimation at the final horizon:

$$\mathcal{E} = \mathbb{E}[(\mu_T - \hat{\mu}_T)^2]$$

Alternatively, another objective, which we develop in the next section, is to minimize the error over the whole trajectory through a regret objective:

$$\mathcal{R}_T = \sum_{t=1}^T \mathbb{E}(\mu_t - \hat{\mu}_t)^2$$

Randomized response We consider local differential privacy, where anyone can access $(Y_{i,t})_{i=1}^{N_t}$, and privacy is ensured at the user level via randomized response (Dwork & Roth, 2013; Ioannidis et al., 2024; Wang et al., 2016). We assume that all users have the same privacy parameter ε and that

this budget remains constant over time. We parameterize by $\beta \in [0, \frac{1}{2}]$ the randomized response:

$$Y_{i,t} \triangleq \begin{cases} X_{i,t}, & \text{with prob. } \frac{1}{2} + \beta \\ 1 - X_{i,t}, & \text{with prob. } \frac{1}{2} - \beta \end{cases}$$

With this definition, the learner's best unbiased estimator is given by

$$\hat{\mu}_t \triangleq \frac{\hat{\nu}_t - \frac{1}{2} + \beta}{2\beta} \quad \text{with} \quad \hat{\nu}_t \triangleq \frac{1}{N_t} \sum_{i=1}^{N_t} Y_{i,t}. \quad (1)$$

Lemma 1. *For any given privacy budget $\varepsilon > 0$, setting the perturbation parameter to*

$$\beta = \frac{1}{2} \tanh\left(\frac{\varepsilon}{2}\right) \quad (2)$$

guarantees that the randomized response mechanism satisfies ε -local differential privacy (ε -LDP).

The proof of this lemma and the derivation of the estimator formula are deferred to Appendix C.

Population evolution The population changes according to two phenomena: new users can join the training, and participating users can leave the training in reaction to data leakage. We define a leakage event as a response remaining unperturbed, i.e., $X_{i,t-1} = Y_{i,t-1}$. In case of data leakage, a participant leaves the training with a fixed probability q , which we assume is independent of time and of the user. This model has the limitation of having no memory – if a user stays despite a leakage event at a given step, they do not become more likely to leave after the next leakage event – but this allows us to minimize the number of model parameters. For the inclusion of new participants, we assume that each participant whose data is not leaked at a given time can recruit an additional participant with probability r . Another possible update rule is to consider growth directly proportional to the total number of users, regardless of whether their data was leaked.¹

Lemma 2. *The evolution of the number of participants is given by:*

$$\mathbb{E}[N_t | N_{t-1}] = N_{t-1} \left(1 - q \left(\frac{1}{2} + \beta \right) + r \left(\frac{1}{2} - \beta \right) \right) \quad (3)$$

We prove this formula in Appendix C.

To analyze the long-term evolution of the user population, it is natural to summarize its expected one-step change via a single growth factor,

$$C(\beta) \triangleq \frac{\mathbb{E}[N_t | N_{t-1}]}{N_{t-1}} = 1 + \frac{r - q}{2} - \beta(q + r). \quad (4)$$

This scalar represents the expected ratio of the population size between two consecutive rounds. The value of $C(\beta)$ relative to 1 dictates the overall trajectory of the system: $C(\beta) \geq 1$ denotes a stable or growing population, whereas $C(\beta) < 1$ indicates decline. By evaluating this one-step expected growth, we can determine the maximum value of β (i.e., the least private the mechanism can be) that the platform can support without seeing the population collapse.

Theorem 1 (Critical Threshold for User Population Survival). *To prevent demographic collapse (i.e., to have $C(\beta) \geq 1$), β must be smaller than the critical threshold β_c , defined as:*

$$\beta_c \triangleq \max \left\{ 0, \frac{r - q}{2(q + r)} \right\} \quad (5)$$

We prove this theorem formally in Appendix C. From this formula, we can see two regimes: if $q > r$, the population will always collapse, regardless of the privacy protection. However, when $r > q$, the population can grow or shrink depending on the choice of β , and thus $C(\beta) \geq 1$ gives a hard operational constraint on the choice of β .

¹We call this the agnostic model and study its behavior in Appendix B.

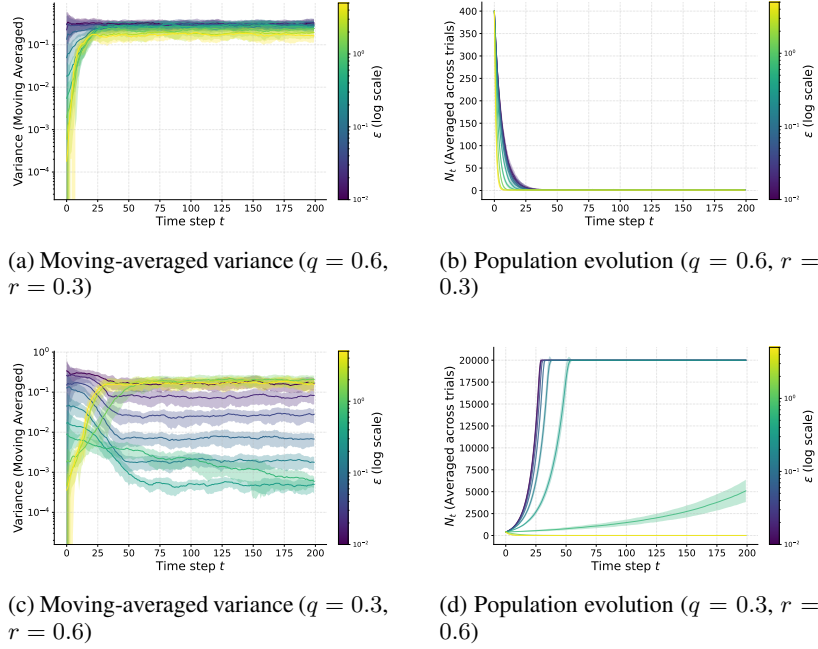


Figure 1: Comparison of moving-averaged variance and population evolution under two parameter settings: the top row corresponds to $(q, r) = (0.6, 0.3)$, while the bottom row corresponds to $(q, r) = (0.3, 0.6)$.

Empirical Simulation We sample binary responses from a Bernoulli distribution with mean μ_t and estimate it using the unbiased estimator equation 1. Responses are privatized by randomized response with perturbation probability $\frac{1}{2} - \beta$, where β is chosen according to equation 2 to satisfy ε -local differential privacy. Utility is measured by the squared estimation error at the final horizon. Throughout, we use a constant privacy budget ε , track the population size N_t (clipped to $[1, 20000]$) and the moving-averaged variance of $\hat{\mu}_t$ (window size 15), and average results over 30 independent trials. Figure 1 illustrates two representative regimes corresponding to $(r, q) = (0.6, 0.3)$ and $(0.3, 0.6)$. For both the variance and population plots, 10 values of ε were used, with their corresponding values indicated by the intensity bar to the left of each plot. As predicted, we observe population collapse for $q > r$, whereas for $r > q$, the population reaches a stable equilibrium or grows to the maximum size, and a finite privacy budget optimally balances privacy noise against user retention. Figure 2 reports the utility-maximizing privacy budget for each (r, q) pair. In the region where a finite optimum exists, the optimal ε decreases monotonically with the leaving rate, confirming that stronger privacy guarantees are required when users are more sensitive to data leakage.

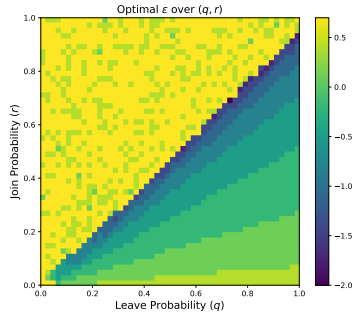


Figure 2: Global mapping of the optimal privacy budget ε^* across the (r, q) parameter space. In the bottom-right region of the figure, the optimal privacy parameter evolves continuously.

4 MEAN ESTIMATION WITH MEMBERSHIP INFERENCE DETECTION

The previous section examined performative population dynamics under randomized response, where data is binary and leakage is defined as the exact release of the true bit. In that setting, utility and privacy risk are strictly opposed, as the leakage is defined as the single bit of information that is shared. This opposition becomes less pronounced for more complex data. We now generalize our framework to continuous vectors in $\mathbb{R}^d$. This transition decouples a record's informational utility from its membership detection; for example, in private mean estimation, a record located exactly at the population mean is statistically difficult to detect, but valuable for the estimation of the mean, while an outlier is easy to detect but does not improve utility. Finally, this high-dimensional continuous setting is closer to realistic machine learning deployments, allowing us to formalize user vulnerability through standard Membership Inference Attacks (MIAs) based on an individual's geometric footprint.

4.1 THE LEARNING TASK AND MECHANISM

At any round t , the user population has size N_t . The platform seeks to estimate the true mean μ of the underlying data distribution $\mathcal{D}$ over $\mathbb{R}^d$. The dataset at round t is denoted as $Z_t = \{z_1, \dots, z_{N_t}\}$, where each individual's record $z_i \sim \mathcal{D}$ is drawn independently. To ensure bounded sensitivity, each record is clipped to a maximum ℓ_2 -norm inside radius R . Thus, $\bar{z}_i = z_i \min\left(1, \frac{R}{\|z_i\|_2}\right)$ denotes the clipped vector z_i . The output o_t is an empirical mean protected by the Gaussian Mechanism:

$$o_t = \mathcal{M}_\gamma(Z_t) = \frac{1}{N_t} \sum_{i=1}^{N_t} \bar{z}_i + \eta_t, \quad \eta_t \sim \mathcal{N}\left(0, \frac{\gamma^2}{N_t} I_{d \times d}\right) \quad (6)$$

This corresponds to each user adding an independent Gaussian noise $\mathcal{N}(0, \gamma^2 I_d)$ to their clipped record (whose ℓ_2 -sensitivity is $2R$). To satisfy a given zero-concentrated differential privacy (zCDP) budget ρ (Bun & Steinke, 2016) for each individual contribution, the platform must set the noise parameter to $\gamma^2 = \frac{2R^2}{\rho}$.

The utility is measured through the expected squared Mahalanobis distance between the private release and the true population mean μ (Biswas et al., 2020; Azize & Basu, 2025; Leemann et al., 2024), which the platform seeks to minimize. The following lemma computes this error explicitly, conditioned on a given population size.

Lemma 3 (Conditional Expected Mahalanobis Error). *Let Z_t be a dataset drawn from a distribution with true mean μ and covariance matrix Σ , and neglect the clipping bias (i.e., assume $\bar{z}_i = z_i$). Since the noise vector η_t has covariance $C_\gamma = \frac{\gamma^2}{N_t} I$, the expected squared Mahalanobis error is:*

$$\mathbb{E} [\|o_t - \mu\|_{\Sigma^{-1}}^2 \mid N_t] = \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_t} \quad (7)$$

In the next Section 4.2, we define the privacy leakage in this setting, and then derive generic bound in Section 4.3 and an exact expression in the Gaussian case in Section 4.4.

4.2 MEMBERSHIP INFERENCE AND THE VULNERABILITY THRESHOLD

Where Section 3 defined leakage via the discrete event $X_{i,t} = Y_{i,t}$, exposure in continuous aggregates ($o_t \in \mathbb{R}^d$) is quantified by a record's statistical influence on o_t . By the Neyman-Pearson lemma, the optimal test statistic for the hypothesis $z \in Z_t$ is the log-likelihood ratio:

$$\Lambda(o_t|z) \triangleq \ln \left(\frac{\Pr[\mathcal{M}_\gamma(Z_t) = o_t \mid z \in Z_t]}{\Pr[\mathcal{M}_\gamma(Z_t \setminus \{z\}) = o_t]} \right) \quad (8)$$

where $\Pr[\cdot]$ denotes the density of the output. The magnitude of $\Lambda(o_t|z)$ indicates how much skew z exerts on the noisy sample mean o_t given we observe the set Z_t .

Vulnerability Threshold (τ). Practically, an individual cannot compute $\Lambda(o_t|z)$ without access to Z_t . However, it is the quantity approximated by external adversaries and auditing tools in modern Membership Inference Attacks (MIAs) [Shokri et al. \(2017\)](#); [Carlini et al. \(2021\)](#). Conditioning user departure on this mathematical quantity thus provides a worst-case upper bound against any adversary. The privacy tolerance is captured via a constant threshold $\tau > 0$, such that a record is compromised if $\Lambda(o_t|z) > \tau$. From a hypothesis testing perspective, maintaining a fixed false positive rate would require scaling the threshold with the dimension d and the population size N_t over time, but we keep it constant for simplicity.

Performative Dynamics. At round t , each participant $z \in Z_t$ updates their status for the subsequent round: if $\Lambda(o_t|z) > \tau$, the individual departs with probability q ; otherwise ($\Lambda(o_t|z) \leq \tau$), the individual remains and recruits one additional participant with probability r . as described in Algorithm 1.

Algorithm 1 Private Mean Estimation under MIA leaving process

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1: Input: Initial size  $N_0$ , Dist.  $\mathcal{D}$ , Mechanism  $\mathcal{M}$ , Threshold  $\tau$ , Probs.  $q, r$ , Rounds  $T$ 
2: Output: Sequence  $(N_1, \dots, N_T)$ 
3: Set  $N_1 = N_0$ 
4: for  $t = 1, \dots, T$  do
5:   if  $N_t = 0$  then
6:     break
7:   Sample fresh dataset  $Z_t \sim \mathcal{D}^{N_t}$ 
8:   Sample candidate output  $o_t \sim \mathcal{M}(Z_t)$  ▷  $\mathcal{M}$  performs  $\ell_2$  clipping  $\forall z \in Z_t$  internally
9:   Initialize next population size  $N_{t+1} \leftarrow N_t$ 
10:  for each  $z \in Z_t$  do ▷ Performative Mechanism
11:     $\Lambda(o_t|z) \leftarrow \ln \left( \frac{\Pr[\mathcal{M}(Z_t)=o_t|z \in Z_t]}{\Pr[\mathcal{M}(Z_t \setminus \{z\})=o_t]} \right)$ 
12:    if  $\Lambda(o_t|z) > \tau$  then
13:       $N_{t+1} \leftarrow N_{t+1} - 1$  with prob.  $q$ 
14:    else
15:       $N_{t+1} \leftarrow N_{t+1} + 1$  with prob.  $r$ 
16: Return  $(N_1, \dots, N_T)$ 

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4.3 GENERAL BOUND ON THE LEAKAGE PROBABILITY

Computing the exact trajectory of the population size requires the knowledge of the marginal leakage probability $p_\tau^{(t)} = \Pr(\Lambda(o_t|z) > \tau)$. For arbitrary data distributions, deriving the exact probability density function of the log-likelihood ratio is intractable. Instead, we provide an upper bound on the leakage probability that involves the Rényi divergence, and thus makes the link with Rényi DP ([Mironov, 2017](#)), via Markov's inequality.

Lemma 4 (Tail Bound on the Leakage Probability). *Let Z_t be the dataset at the beginning of round t and $z \in Z_t$ be a target record. Let $o_t \sim \mathcal{M}(Z_t)$ be the candidate output. Define the factual and counterfactual output distributions as $P_{in}^{\mathcal{M}}(\cdot) = \Pr[\mathcal{M}(Z_t) = \cdot \mid z \in Z_t]$ and $P_{out}^{\mathcal{M}}(\cdot) = \Pr[\mathcal{M}(Z_t \setminus \{z\}) = \cdot]$.*

For the log-likelihood ratio $\Lambda(o_t|z) = \ln \left(\frac{P_{in}^{\mathcal{M}}(o_t)}{P_{out}^{\mathcal{M}}(o_t)} \right)$ evaluated at output o_t , and for any threshold $\tau > 0$ and order $\alpha > 1$, the tail probability of leakage is bounded by:

$$\Pr_{o_t \sim P_{in}^{\mathcal{M}}}(\Lambda(o_t|z) > \tau) \leq \exp \left(-(\alpha - 1)(\tau - D_\alpha(P_{in}^{\mathcal{M}} \parallel P_{out}^{\mathcal{M}})) \right)$$

where $D_\alpha(P_{in}^{\mathcal{M}} \parallel P_{out}^{\mathcal{M}})$ is the Rényi divergence of order α .

The bound in 4 can easily be applied to any DP mechanism outputs, as it depends explicitly on the Rényi divergence. It also exhibits exponential dependency in τ , demonstrating the threshold carefully. Finally, to make the tightest possible bound, we should optimize over the order α while keeping it strictly greater than 1 to use RDP. To illustrate it, we apply it to the Gaussian mechanism. However, for analytical tractability we condition the leakage probability on observing Z_t first.

4.4 THE GAUSSIAN PARTICULAR CASE

In this subsection, we exploit the fact that the injected algorithmic noise is normally distributed to derive the exact closed-form distribution of the log-likelihood ratio conditioned on the data for any given point. We then derive an exact closed-form expression for the unconditional leakage probability under the additional assumption that the data distribution is Gaussian. This Gaussian-data assumption is required only for the results beginning with lemma 8 and is not needed for the preceding results. For simplicity, we initially assume that $\mathcal{M}$ is clipping-free. This assumption will be relaxed in subsequent results. Therefore, let $\mu_z \triangleq \frac{1}{N_t} \sum_{i=1}^{N_t} z_i$ denote the sample mean of z_i 's.

Lemma 5 (Upper Bound for the Gaussian Mechanism). *Substituting into Lemma 4 the Rényi divergence of the Gaussian mechanism equation 6 between the outputs with and without the record z conditioned on the entire dataset Z_t , namely*

$$D_\alpha(P_{in}^{\mathcal{M}}(\cdot|Z_t) \parallel P_{out}^{\mathcal{M}}(\cdot|Z_t)) = \frac{\alpha N_t}{2\gamma^2(N_t - 1)^2} \|z - \mu_z\|^2$$

(see Lemma 11 in Appendix D), the tail bound on the leakage probability for a specific record z specializes to:

$$\Pr_{o_t \sim P_{in}^{\mathcal{M}}}(\Lambda(o_t|z) > \tau \mid Z_t) \leq \exp\left(-(\alpha - 1) \left(\tau - \frac{\alpha N_t \|z - \mu_z\|^2}{2\gamma^2(N_t - 1)^2}\right)\right). \quad (9)$$

This closed-form bound makes explicit, first, the dependence on $\|z - \mu_z\|^2$: outliers face an exponentially increasing probability of leakage as their deviation from the empirical mean grows. Second, when the population size N_t increases, the divergence decreases, capturing the privacy amplification afforded by aggregation and leading to a virtuous circle: if the system is in a regime private enough to gain users, it becomes more likely to keep growing as the leakage risk decreases.

This inequality can be minimized with respect to the Rényi order α , yielding the optimal order

$$\alpha^* = \frac{1}{2} + \frac{\tau\gamma^2(N_t - 1)^2}{N_t\|z - \mu_z\|^2} \quad \text{subject to } \alpha^* > 1.$$

Thus the optimized bound is valid and non-vacuous only when

$$\tau > \frac{N_t\|z - \mu_z\|^2}{2\gamma^2(N_t - 1)^2},$$

i.e., only when the individual's baseline privacy tolerance strictly exceeds their deviation from the empirical mean rescaled by the current parameters. This condition fails precisely for extreme spatial outliers and in low-noise regimes ($\gamma \rightarrow 0$), where $\alpha^* \leq 1$ either invalidates the divergence or trivializes the bound (i.e., it collapses to 1). Writing

$$a \triangleq \frac{N_t\|z - \mu_z\|^2}{2\gamma^2(N_t - 1)^2},$$

substituting α^* back into equation 9 yields, for $\tau > a$, the optimized bound

$$\Pr_{o_t \sim P_{in}^{\mathcal{M}}}(\Lambda(o_t|z) > \tau \mid Z_t) \leq \exp\left(-\frac{(\tau - a)^2}{4a}\right). \quad (10)$$

As Lemma 6 below shows that

$$\Lambda(o_t|z) \mid Z_t \sim \mathcal{N}(a, 2a)$$

exactly, the standard Gaussian tail bound $1 - \Phi(u) \leq e^{-u^2/2}$ applied at $u = (\tau - a)/\sqrt{2a}$ gives the same exponent: the optimized Rényi bound is tight up to the Φ -versus-exp gap. In the special case of the Gaussian mechanism, we are however able to provide the exact leakage probability through a direct computation of Λ .

Lemma 6. *Let a mechanism output o_t be perturbed by additive isotropic Gaussian noise $\eta_t \sim \mathcal{N}(0, \frac{\gamma^2}{N_t} I_{d \times d})$, representing the noise. The log-likelihood ratio $\Lambda(o_t|z)$ conditioned on Z_t is given by:*

$$\Lambda(o_t|z) \mid Z_t = \frac{N_t\|z - \mu_z\|^2}{2(N_t - 1)^2\gamma^2} + \frac{N_t\langle z - \mu_z, \eta_t \rangle}{(N_t - 1)\gamma^2}$$

And it is distributed as:

$$\Lambda(o_t|z) \mid Z_t \sim \mathcal{N}\left(\frac{N_t\|z - \mu_z\|^2}{2(N_t - 1)^2\gamma^2}, \frac{N_t\|z - \mu_z\|^2}{(N_t - 1)^2\gamma^2}\right)$$

Lemma 7. *Given the aggregated empirical mean mechanism, the exact probability that a specific individual data point z is purged (i.e., its log-likelihood ratio breaches the threshold τ) conditioned on Z_t is given in closed form by:*

$$\Pr(\Lambda(o_t|z) > \tau \mid Z_t) = 1 - \Phi\left(\frac{\tau(N_t - 1)\gamma}{\sqrt{N_t}\|z - \mu_z\|} - \frac{\sqrt{N_t}\|z - \mu_z\|}{2(N_t - 1)\gamma}\right)$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

Unlike the generalized Rényi divergence bound in equation 9, which yields a conservative upper limit subject to rigid parametric constraints, Lemma 7 exactly characterizes the tail probability of leakage. This closed-form expression eliminates the theoretical fragility associated with spatial outliers.

Remark 1 (Connection to Gaussian DP). With $\mu_{z'} \triangleq \sqrt{N_t}\|z - \mu_z\|/(\gamma(N_t - 1)) \approx \|z - \mu_z\|/(\gamma\sqrt{N_t})$, Lemma 6 reads $\Lambda(o_t|z) \mid Z_t \sim \mathcal{N}(\mu_{z'}^2/2, \mu_{z'}^2)$, which is the privacy loss of a $\mu_{z'}$ -Gaussian-DP pair (Dong et al., 2022): for the record z , membership inference is exactly as hard as testing $\mathcal{N}(0, 1)$ against $\mathcal{N}(\mu_{z'}, 1)$. Lemma 7 is then the power of the Neyman–Pearson test at likelihood-ratio threshold e^τ — equivalently, of the Bayes-optimal attack with prior odds of membership $e^{-\tau}$. Thus, maintaining a fixed FPR ν corresponds to the record-dependent threshold $\tau = \mu_{z'}\Phi^{-1}(1 - \nu) - \mu_{z'}^2/2$, as discussed in Section 4.2, and since $\|\bar{z}\|_2, \|\mu_{\bar{z}}\|_2 \leq R$, one recovers the worst-case $(2R/(\gamma\sqrt{N_t}))$ -Gaussian-DP guarantee.

We now focus on this setting allowing the clipping of vectors for the rest of our results, under the following assumption.

Assumption 1. *We assume that the records are drawn independently and identically from a d -dimensional multivariate Gaussian distribution, ($z \sim \mathcal{N}(\mu, \Sigma)$), with probability density function*

$$f_{\mathcal{N}}(z) = \frac{1}{\sqrt{(2\pi)^d|\Sigma|}} \exp\left(-\frac{1}{2}(z - \mu)^T \Sigma^{-1}(z - \mu)\right).$$

Lemma 8 (Expected Population Evolution). *We define the marginal leakage probability, $p_\tau^{(t-1)}$, as the expected fraction of the population that breaches the privacy threshold τ at round $t - 1$, i.e. $p_\tau^{(t-1)} = \mathbb{E}(\mathbb{1}(\Lambda(o_{t-1}|z) > \tau))$. Letting $\bar{z}$ denote the record subjected to the clipping and $\mu_{\bar{z}}$ denote the sample mean of clipped vectors, the marginal leakage probability evaluates to:*

$$p_\tau^{(t-1)} = \int_{\mathbb{R}^{dN_{t-1}}} \left[1 - \Phi\left(\frac{\tau(N_{t-1} - 1)\gamma}{s\sqrt{N_{t-1}}} - \frac{s\sqrt{N_{t-1}}}{2(N_{t-1} - 1)\gamma}\right)\right] \prod_{j=1}^{N_{t-1}} f_{\mathcal{N}}(z_j) dz_{1:N_{t-1}}, \quad (11)$$

Where $s \triangleq \|\bar{z} - \mu_{\bar{z}}\|_2$ and $dz_{1:N_{t-1}} \triangleq dz_1 \dots dz_{N_{t-1}}$. Under assumption 1, the expected population size at round t in Algorithm 1, is given by:

$$\mathbb{E}[N_t \mid N_{t-1}] = N_{t-1} \left(1 + r - (q + r)p_\tau^{(t-1)}\right) \quad (12)$$

Lemma 8 gives an exact characterization of the user population evolution. As in Lemma 2, the expected population is the previous population multiplied by a growth factor. However, unlike the previous case, this factor is determined by the marginal leakage probability p_τ , whose integral form captures the dependence on several parameters: the dimension d , the threshold τ , the clipping bound R and the noise scale γ , instead of relying only on the privacy parameter β .

4.5 CRITICAL NOISE THRESHOLD FOR USER POPULATION SURVIVAL

To determine whether the user population survives or collapses, we denote similarly as before $C_t(\gamma)$ the expected ratio of the population size between consecutive rounds t and $t - 1$:

$$C_t(\gamma) \triangleq \frac{\mathbb{E}[N_t \mid N_{t-1}]}{N_{t-1}} = 1 + r - (q + r)p_\tau^{(t-1)}(\gamma) \quad (13)$$

The continuous growth factor $C_t(\gamma) = 1 + r - (q+r)p_\tau^{(t-1)}(\gamma)$ mirrors the discrete model of Lemma 2 by substituting the leakage probability $\frac{1}{2} + \beta$ of randomized response with the marginal leakage probability $p_\tau^{(t-1)}(\gamma)$. This substitution preserves the system’s monotonicity: increasing the noise γ strictly decreases $p_\tau^{(t-1)}(\gamma)$, improving population stability. However, because $p_\tau^{(t-1)}(\gamma)$ requires evaluating a non-linear integral over a multivariate data distribution, the boundary condition $C_t(\gamma) \geq 1$ cannot be inverted analytically in general, but it can easily be computed numerically. Moreover, C is now time-dependent unlike in section 3, introducing an additional layer of performative dynamics into the system.

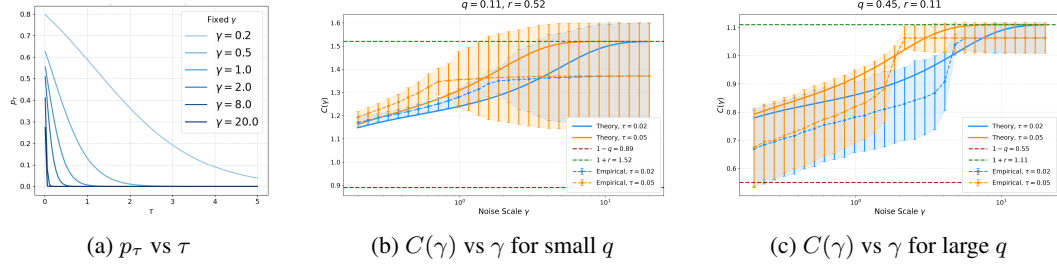


Figure 3: Behavior of p_τ with τ and $C(\gamma)$ with γ , comparing theoretical predictions with empirical simulations. (a) $N = 100$; (b) $T = 50$; (c) $T = 10$, with the population clipped at 20,000.

Since the marginal leakage probability $p_\tau(\gamma)$ is a function of the mechanism’s noise scale, the platform can directly steer the demographic growth by tuning γ . The boundary condition for demographic survival is $C(\gamma) \geq 1$. Under the minor assumption that the clipping bound $R = \Omega(\sqrt{d})$, the following theorem establishes a critical noise threshold γ_c that is sufficient to satisfy this condition and prevent a demographic collapse.

Theorem 2 (Critical Noise Threshold for Demographic Survival). *The condition $C_t(\gamma) \geq 1$ is guaranteed as soon as the mechanism’s noise scale satisfies the explicit lower bound $\gamma \geq \gamma_c$. Given an ℓ_2 clipping bound $R = \Omega(\sqrt{d})$, the critical threshold γ_c is defined as:*

$$\gamma_c \triangleq \Omega \left(\frac{\sqrt{dN_{t-1}}}{N_{t-1} - 1} \left(\frac{\Phi_{q,r}^{-1} + \sqrt{(\Phi_{q,r}^{-1})^2 + 2\tau}}{2\tau} \right) \right) \quad (14)$$

where $\Phi_{q,r}^{-1} \triangleq \Phi^{-1} \left(\frac{q}{q+r} \right)$ is the probit function (the inverse of the standard normal cumulative distribution function) evaluated at the performative survival threshold.

As before, $C_t(\gamma) < 1$ implies a demographic collapse initiated at round t , destroying the platform’s utility over time, which means that a minimum level of privacy is needed for long-term success. Furthermore, the explicit formulation of γ_c decouples the geometric requirements of the data distribution ($\approx \sqrt{d/N_{t-1}}$) from the purely behavioral constraints of the population (the $\Phi_{q,r}^{-1}$ and τ terms).

Recall that in this setting the utility is measured by the Mahalanobis error; the following theorem provides tight, finite-sample bounds on its expectation.

Theorem 3 (Finite-Sample Bounds on the Expected Mahalanobis Error). *Assume that the population is floored at one participant ($N_t \geq 1$). Conditioned on N_{t-1} , the expected squared Mahalanobis error of the private empirical mean $\mathbf{o}_t$ satisfies:*

$$\frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{C_t(\gamma)N_{t-1}} \leq \mathbb{E} [\|\mathbf{o}_t - \mu\|_{\Sigma^{-1}}^2 \mid N_{t-1}] \leq \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{C_t(\gamma)N_{t-1}} \cdot \mathcal{O} \left(\frac{1}{1-q} \right) \quad (15)$$

where the asymptotic notation in the upper bound is particularly useful in the regime $q \rightarrow 1$.

Proof sketch. Since the estimation error scales as $1/N_t$, we bound $\mathbb{E}[1/N_t \mid N_{t-1}]$. The lower bound follows from Jensen’s inequality, while the upper bound uses the Kantorovich-type inequality for the multiplicative inverse function; see Appendix D. $\square$

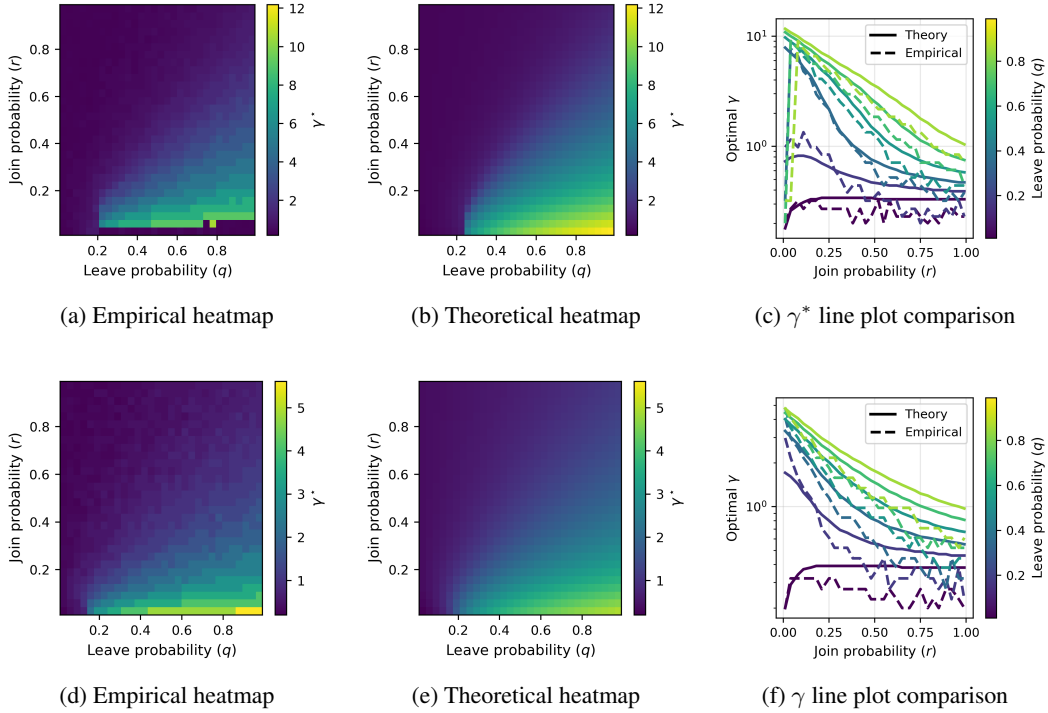


Figure 4: Comparison of the optimal noise scale γ^* obtained from theory and simulations. The first row corresponds to $\tau = 0.02$, while the second row corresponds to $\tau = 0.05$. For each value of τ , the left panel shows the empirical heatmap, the middle panel shows the theoretical heatmap, and the right panel compares the theoretical and empirical optimal γ as a function of the join probability.

Theorem 3 establishes that the expected unconditional estimation error at round t , i.e., $\mathcal{E}_t(\gamma) \triangleq \mathbb{E}[|o_t - \mu|_{\Sigma^{-1}}^2]$, balances the mechanism variance in the numerator against the expected population size, $\mathbb{E}[N_t] = \mathbb{E}[\mathbb{E}[N_t | N_{t-1}]]$, in the denominator. Using a mean-field approximation on N_t , we approximate the latter as $\mathbb{E}[N_t] \approx N_0 \prod_{k=1}^t C_k(\gamma)$. Since we keep γ constant throughout the entire experiment, we need to scale γ_c with the minimum population size permissible by the system, say $N_{\min}$ to ensure $C_t(\gamma) > 1$ for all t , i.e., $\gamma_c \propto 1/\sqrt{N_{\min}}$. This explicit scaling law allows us to evaluate the system's asymptotic behavior under extreme noise regimes.

Remark 2 (Finite-Horizon Optimization). Corollaries 1 and 2 prove that the zero-noise regime ($\gamma \rightarrow 0^+$) implies geometric demographic collapse and diverging error. On the other hand, when noise increases ($\gamma \rightarrow \infty$) it enables unbounded exponential growth that eventually dominates the variance injected into the mechanism in our simplified model. Thus, for an infinite time horizon, infinite noise is theoretically optimal: the more private the mechanism is, the faster the growth. As practitioners operate under a finite horizon T where the total population remains strictly bounded, we complement this result with a finite-horizon analysis.

Lemma 9 (Upper Bound on Cumulative Expected Error over a Finite Horizon). For a finite operational timeline T and an initial population N_0 , let $C_t(\gamma) > 1$ denote the strictly positive expected per-round demographic growth factor for all t . Under mean field-approximation of N_t , the cumulative expected Mahalanobis error aggregated across all T rounds, defined as $\mathcal{E}_{\text{total}}(\gamma) \triangleq \sum_{t=1}^T \mathcal{E}_t(\gamma)$ is upper bounded by $\mathcal{E}_{\text{total}}(\gamma) \leq \hat{\mathcal{E}}_{\text{total}}(\gamma)$, where:

$$\hat{\mathcal{E}}_{\text{total}}(\gamma) \triangleq \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0(C_1(\gamma) - 1)} \left(1 - \frac{1}{[C_1(\gamma)]^T} \right) \quad (16)$$

Remark 3 (Optimization in the Growth Regime). In the growth regime, where $\gamma \geq \gamma_c$ and consequently $C_t(\gamma) > 1$ for all t , directly optimizing the upper bound $\hat{\mathcal{E}}_{\text{total}}(\gamma)$ rather than the exact cumulative error $\mathcal{E}_{\text{total}}(\gamma)$ is a principled and practical surrogate objective. Corollary 2 shows that $\mathcal{E}$ and $\hat{\mathcal{E}}$ share the same convergence rate. Computationally, this substitution is highly advantageous

as it depends solely on the initial expected growth factor $C_1(\gamma)$, bypassing the need to recursively track the dynamic sequence $C_t(\gamma)$ across all T rounds. We leverage this in Theorem 4 (Appendix E) to establish a PAC bound on the Mahalanobis error $\|\alpha_t - \mu\|_{\Sigma^{-1}}$ for the growth regime $\gamma \geq \gamma_c$ and consequently derive a high probability lower bound for t . Furthermore, empirical evaluations confirm that optimizing this upper bound yields a highly effective configuration in practice.

Optimization of the Global Noise Scale (γ^*) This closed-form objective in equation 16 makes explicit the trade-off between providing better utility at a given round and a larger population at the next round. Increasing γ penalizes the numerator via the quadratic variance term (γ^2), but simultaneously reduces the error by increasing the growth factor $C(\gamma)$ in the denominator. We cannot compute a closed form of $\gamma^* = \arg \min_{\gamma > \gamma_c} \hat{\mathcal{E}}_{\text{total}}(\gamma)$. Indeed, the marginal leakage $p_\tau^{(0)}$ nested within $C_1(\gamma)$ is governed by the standard normal cumulative distribution function $\Phi(\cdot)$, so setting the derivative of this objective to zero does not yield a closed-form algebraic solution. However, $\hat{\mathcal{E}}_{\text{total}}(\gamma)$ is a smooth, one-dimensional objective function and γ^* can easily be computed with numerical solvers.

We thus compute the theoretical γ^* heatmap over the population’s recruitment rate r and leaving rate q . As reported in Figure 4 (and also in Figure 5 in Appendix A), the theoretical γ^* closely aligns with the optimal empirical parameters discovered via grid search.

5 DISCUSSION AND FUTURE WORK

We showed that private mean estimation can outperform non-private estimation in the long term when participants drop out after their data is leaked. These results demonstrate that utility maximization can align with strong privacy protection when there is a sufficiently strong feedback loop. This paper introduces performative privacy as the study of the long-term interplay between privacy and utility under repeated model deployment, at the intersection of privacy-preserving machine learning and performative learning. We focused on mean estimation, but we plan to extend our results to more complex learning tasks. The feedback loop is also kept as simple as possible, but could be made more realistic using epidemic models and more complex privacy preferences and composition over time. Finally, this could motivate varying the privacy budget over time, in particular to minimize regret over the full trajectory.

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Appendix

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A MORE EXPERIMENTS ON MEMBERSHIP INFERENCE LEAVING PROCESS

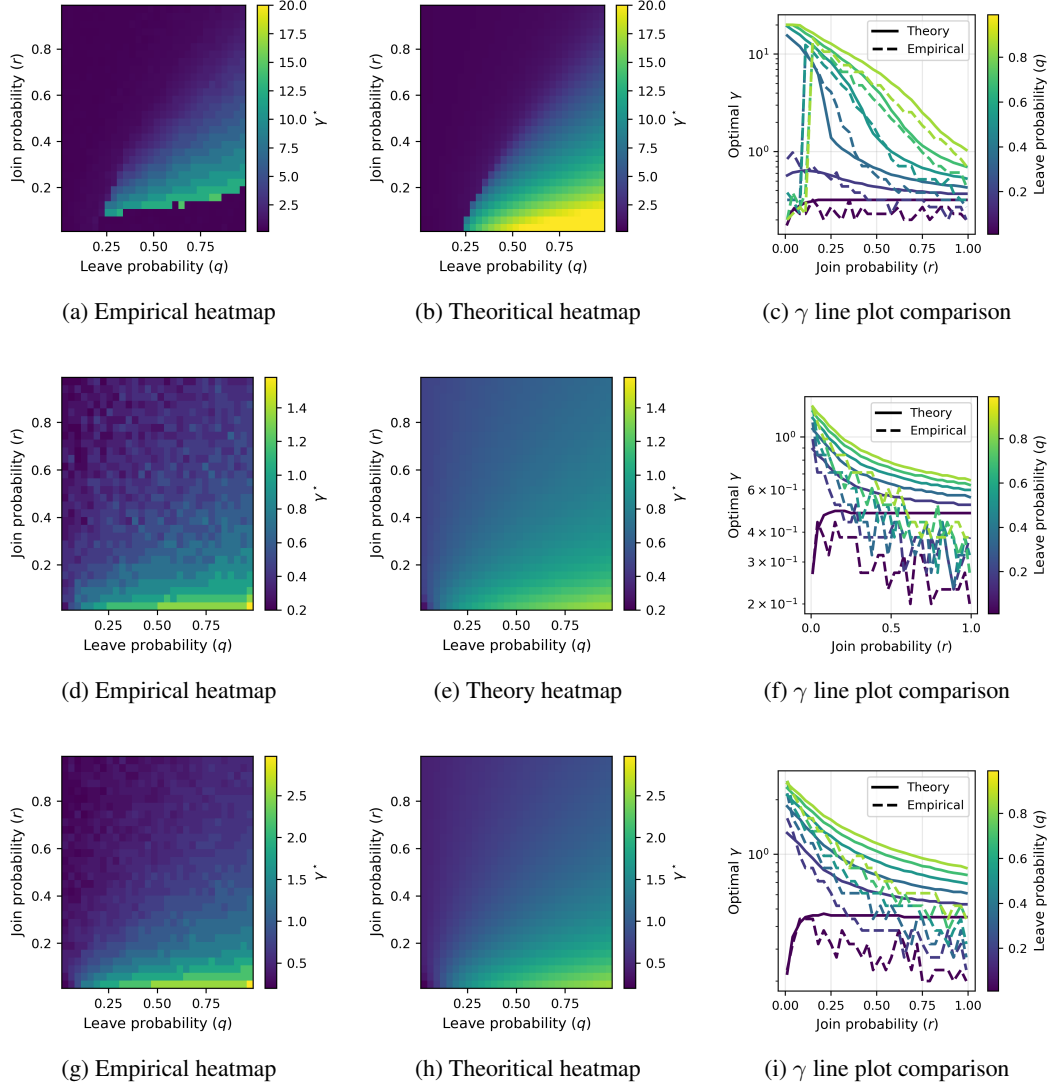


Figure 5: Comparison of the optimal threshold γ obtained from theory and simulations. The first row corresponds to $\tau = 0.01$, the second row corresponds to $\tau = 0.2$ and the third row corresponds to $\tau = 0.1$. For each value of τ , the left panel shows the empirical heatmap, the middle panel shows the theoretical heatmap, and the right panel compares the theoretical and empirical optimal γ as a function of the join probability.

Figures 4 and 5 reveal an interesting observation: as (τ) increases, the empirical noise increasingly underestimates the theoretical prediction because leakage becomes more sensitive to rare, extreme outliers. While the theoretical model assumes an unbounded population, the experimental setup imposes a population cap of 20,000, limiting the occurrence of such extreme samples. Consequently, the discrepancy between theory and experiment becomes more pronounced at larger (τ).

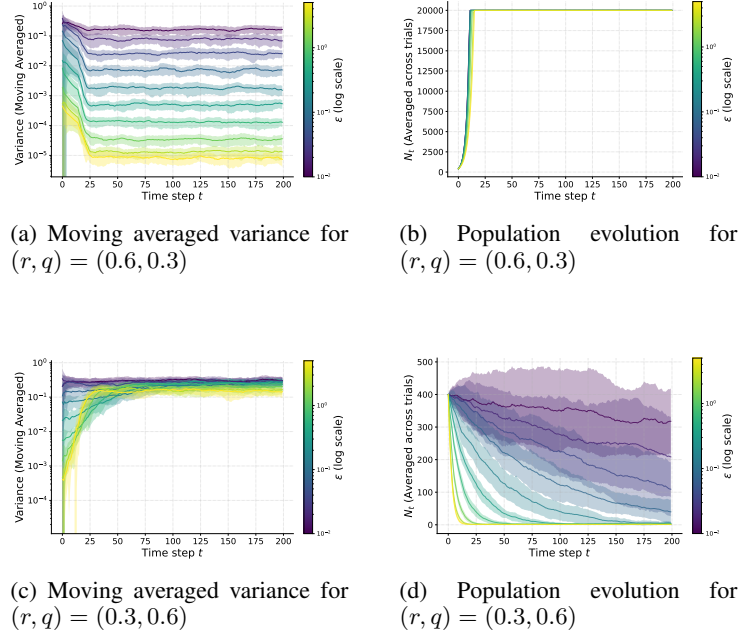


Figure 6: In a growing population ($r = 0.6, q = 0.3$), the system quickly reaches capacity across all privacy budgets, causing variance to stratify based on noise levels and favoring low privacy due to the high growth rate. In contrast, a shrinking population ($r = 0.3, q = 0.6$) triggers performative collapse; while weak privacy (high ε) initially yields higher utility, it accelerates population decline. This causes variance to spike and eventually converge with stricter privacy regimes as the disappearing user base eliminates any initial advantage.

B THE AGNOSTIC MODEL

This section explores the agnostic model mentioned in Section 3. Under this framework, we assume population growth is directly proportional to the number of participants in the previous iteration, $t - 1$, regardless of whether or not their data were leaked. Applying a derivation analogous to the proof of Lemma 2, we characterize the evolution of the participant pool as follows:

$$\begin{aligned}
 N_t &= N_{t-1} - N_{t-1}q \left(\frac{1}{2} + \beta \right) + rN_{t-1} \\
 &= N_{t-1} \left(1 - q \left(\frac{1}{2} + \beta \right) + r \right)
 \end{aligned}$$

Under this model, we once again simulate the behavior of the population and variance under the same two regimes of $(r, q) = (0.3, 0.6)$ and $(r, q) = (0.6, 0.3)$ respectively as demonstrated in Figure 6. In the regime ($r > q$), rapid growth outweighs attrition, driving the population to maximum capacity regardless of the privacy budget. Unlike the previous model — where low privacy could stifle growth or lead to suboptimal stability, this model provides no incentive for privacy, as the highest ε consistently yields the lowest variance. Conversely, when $q > r$, the variance mirrors the original model’s behavior, yet population decay is notably more gradual, with certain ε values inducing high-variance stochastic collapse across different runs.

We also report the utility-maximizing privacy budget ε for each parameter pair in Figure 7. For growing populations ($r > q$), the models diverge sharply: the original model shows a broad transition favoring intermediate privacy, whereas the agnostic model overwhelmingly favors minimal privacy (highest ε), as rapid growth drives the population to capacity regardless of data use. Maximum privacy is beneficial only within a narrow band just below $r = q$; outside this regime, strict privacy provides no utility gain.

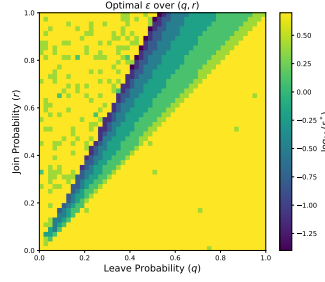


Figure 7: Global mapping of the optimal privacy budget ε^* across the (r, q) parameter space for the agnostic model

C PROOFS FROM SECTION 3

C.1 PROPERTIES OF THE ESTIMATOR

Lemma 10. *The estimator $\hat{\mu}_t$ for population Bernoulli mean μ_t as described in equation 1 is unbiased.*

Proof. To prove that the estimator $\hat{\mu}_t$ is unbiased, we first find the expected value of a single privatized response $Y_{i,t}$. As $X_{i,t}$ is a Bernoulli variable with mean μ_t , we have:

$$\begin{aligned}
 \mathbb{E}[Y_{i,t}] &= \mathbb{P}(X_{i,t} = 1)\mathbb{E}[Y_{i,t} \mid X_{i,t} = 1] + \mathbb{P}(X_{i,t} = 0)\mathbb{E}[Y_{i,t} \mid X_{i,t} = 0] \\
 &= \mu_t \left(\frac{1}{2} + \beta \right) + (1 - \mu_t) \left(\frac{1}{2} - \beta \right) \\
 &= \mu_t \left(\frac{1}{2} + \beta \right) + \frac{1}{2} - \beta - \mu_t \left(\frac{1}{2} - \beta \right) \\
 &= \mu_t \left(\frac{1}{2} + \beta - \frac{1}{2} + \beta \right) + \frac{1}{2} - \beta \\
 &= 2\beta\mu_t + \frac{1}{2} - \beta
 \end{aligned}$$

Now we evaluate the expected value of the sample mean $\hat{\nu}_t$. By linearity of expectation:

$$\begin{aligned}
 \mathbb{E}[\hat{\nu}_t] &= \mathbb{E} \left[\frac{1}{N_t} \sum_{i=1}^{N_t} Y_{i,t} \right] = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbb{E}[Y_{i,t}] \\
 &= \frac{1}{N_t} \cdot N_t \left(2\beta\mu_t + \frac{1}{2} - \beta \right) = 2\beta\mu_t + \frac{1}{2} - \beta
 \end{aligned}$$

Substituting $\mathbb{E}[\hat{\nu}_t]$ into the expected value of our estimator $\hat{\mu}_t$:

$$\begin{aligned}
 \mathbb{E}[\hat{\mu}_t] &= \mathbb{E} \left[\frac{\hat{\nu}_t - \frac{1}{2} + \beta}{2\beta} \right] = \frac{\mathbb{E}[\hat{\nu}_t] - \frac{1}{2} + \beta}{2\beta} \\
 &= \frac{(2\beta\mu_t + \frac{1}{2} - \beta) - \frac{1}{2} + \beta}{2\beta} \\
 &= \frac{2\beta\mu_t}{2\beta} = \mu_t
 \end{aligned}$$

This concludes the proof. □

Lemma 1. *For any given privacy budget $\varepsilon > 0$, setting the perturbation parameter to*

$$\beta = \frac{1}{2} \tanh \left(\frac{\varepsilon}{2} \right)$$

guarantees that the randomised response mechanism satisfies ε -local differential privacy (ε -LDP).

Proof. To satisfy ε -LDP, the maximum ratio of output probabilities (true reporting vs. false reporting) must equal e^ε :

$$\frac{\frac{1}{2} + \beta}{\frac{1}{2} - \beta} = e^\varepsilon$$

Solving directly for β gives:

$$\beta = \frac{1}{2} \left(\frac{e^\varepsilon - 1}{e^\varepsilon + 1} \right)$$

Using the identity $\tanh(z) = \frac{e^{2z} - 1}{e^{2z} + 1}$ with $z = \frac{\varepsilon}{2}$, this simplifies to:

$$\beta = \frac{1}{2} \tanh \left(\frac{\varepsilon}{2} \right)$$

□

C.2 POPULATION EVOLUTION AND CRITICAL NOISE THRESHOLD

Lemma 2. *The evolution of the number of participants is given by:*

$$\begin{aligned} \mathbb{E}[N_t | N_{t-1}] &= N_{t-1} - N_{t-1}q \left(\frac{1}{2} + \beta \right) + r \left(\frac{1}{2} - \beta \right) N_{t-1} \\ &= N_{t-1} \left(1 - q \left(\frac{1}{2} + \beta \right) + r \left(\frac{1}{2} - \beta \right) \right) \end{aligned}$$

Proof. Let D_t and A_t be the number of departures and arrivals at t . The population at time t follows the basic balance equation:

$$N_t = N_{t-1} - D_t + A_t$$

By construction of the model, A_t and D_t follow $\text{Binomial}(N_{t-1}, r(1/2 - \beta))$ and $\text{Binomial}(N_{t-1}, q(\beta + 1/2))$ respectively. For a sufficiently large population, by the Law of Large Numbers, A_t and D_t tightly concentrate around their expected values (first-order approximation).

Thus, $\mathbb{E}[A_t | N_{t-1}] = r \left(\frac{1}{2} - \beta \right) N_{t-1}$ and $\mathbb{E}[D_t | N_{t-1}] = N_{t-1}q \left(\frac{1}{2} + \beta \right)$. Substituting these deterministic expectations into the balance equation yields:

$$\begin{aligned} \mathbb{E}[N_t | N_{t-1}] &= N_{t-1} - N_{t-1}q \left(\frac{1}{2} + \beta \right) + r \left(\frac{1}{2} - \beta \right) N_{t-1} \\ &= N_{t-1} \left(1 - q \left(\frac{1}{2} + \beta \right) + r \left(\frac{1}{2} - \beta \right) \right) \end{aligned}$$

□

Theorem 1. *To maintain $C(\beta) \geq 1$, the parameter β must satisfy the explicit upper bound $\beta \leq \beta_c$, where the critical threshold β_c is defined as:*

$$\beta_c \triangleq \max \left\{ 0, \frac{r - q}{2(q + r)} \right\}$$

Proof. Substituting the simplified linear expression for the performative growth factor yields:

$$\begin{aligned} 1 + \frac{r - q}{2} - \beta(q + r) &\geq 1 \\ \implies \frac{r - q}{2} - \beta(q + r) &\geq 0 \end{aligned}$$

Rearranging the inequality to isolate the β term yields:

$$\beta(q + r) \leq \frac{r - q}{2}$$

Dividing both sides by $(q + r)$,

$$\beta \leq \frac{r - q}{2(q + r)}$$

However, when $r \leq q$, the critical point get clipped to zero. Thus,

$$\beta \leq \max \left\{ 0, \frac{r - q}{2(r + q)} \right\}$$

This completes the proof. $\square$

D PROOFS FROM SECTION 4

D.1 GENERIC RESULTS

Lemma 3. *Let Z_t be a dataset drawn from a distribution with true mean μ and covariance matrix Σ , and neglect the clipping bias (i.e., assume $\tilde{z}_i = z_i$). Assuming the aggregated noise vector η_d has covariance $C_\gamma = \frac{\gamma^2}{N_t} I$, the expected squared Mahalanobis error is:*

$$\mathbb{E} [\|o_t - \mu\|_{\Sigma^{-1}}^2 \mid N_t] = \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_t} \quad (17)$$

Proof. Since the mechanism's injected noise η_d is sampled independently from the underlying data distribution $\mathcal{D}$, the expected squared error decouples into two components: the variance introduced by the privacy mechanism and the intrinsic geometric variance of the data sample itself.

By applying the linearity of expectation and utilizing the trace trick for quadratic forms ($\mathbb{E}[x^T A x] = \text{Tr}(A \Sigma_x) + \mu_x^T A \mu_x$), we expand the conditional error:

$$\begin{aligned} \mathbb{E} [\|o_t - \mu\|_{\Sigma^{-1}}^2 \mid N_t] &= \mathbb{E}_{\eta_d} [\|\eta_d\|_{\Sigma^{-1}}^2] + \frac{1}{N_t^2} \mathbb{E}_{\mathcal{D}} \left[\left\| \sum_{z_i \in Z_t} (z_i - \mu) \right\|_{\Sigma^{-1}}^2 \right] \\ &= \text{Tr}(\Sigma^{-1} C_\gamma) + \frac{1}{N_t^2} \sum_{z_i \in Z_t} \mathbb{E}_{\mathcal{D}} [\|z_i - \mu\|_{\Sigma^{-1}}^2] \\ &= \frac{\gamma^2 \text{Tr}(\Sigma^{-1})}{N_t} + \frac{1}{N_t^2} \sum_{z_i \in Z_t} \text{Tr}(\Sigma^{-1} \Sigma) \\ &= \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_t} \end{aligned}$$

where we substitute $C_\gamma = \gamma^2 I$ and recognize that $\text{Tr}(\Sigma^{-1} \Sigma) = \text{Tr}(I_d) = d$. This bounds the conditional expected error, completing the proof. $\square$

Lemma 4. *Let Z_{t-1} be the dataset at round t and $z \in Z_t$ be a target record. Let $o_t \sim \mathcal{M}(Z_t)$ be the candidate output. Define the factual and counterfactual output distributions as $P_{in}^{\mathcal{M}}(\cdot) = \Pr[\mathcal{M}(Z_t) = \cdot \mid z \in Z_t]$ and $P_{out}^{\mathcal{M}}(\cdot) = \Pr[\mathcal{M}(Z_t \setminus \{z\}) = \cdot]$.*

For the log-likelihood ratio $\Lambda(o_t|z) = \ln \left(\frac{P_{in}^{\mathcal{M}}(o_t)}{P_{out}^{\mathcal{M}}(o_t)} \right)$ evaluated at output o_t , and for any threshold $\tau > 0$ and order $\alpha > 1$, the tail probability of structural compromise is bounded by:

$$\Pr_{o_t \sim P_{in}^{\mathcal{M}}} (\Lambda(o_t|z) > \tau) \leq \exp \left(-(\alpha - 1)(\tau - D_\alpha(P_{in}^{\mathcal{M}} \parallel P_{out}^{\mathcal{M}})) \right)$$

where $D_\alpha(P_{in}^{\mathcal{M}} \parallel P_{out}^{\mathcal{M}})$ is the Rényi divergence of order α .

Proof. We use the Cramer-Chernoff technique to bound the leakage probability of z . By the monotonicity of the exponential function, for any $\alpha > 1$, the tail event $\Lambda(o_t|z) > \tau$ is strictly equivalent to:

$$e^{(\alpha-1)\Lambda(o_t|z)} > e^{(\alpha-1)\tau}$$

Applying Markov's inequality to the non-negative random variable $e^{(\alpha-1)\Lambda(o_t|z)}$ over the factual distribution $P_{\text{in}}^{\mathcal{M}}$ yields:

$$\Pr_{o_t \sim P_{\text{in}}^{\mathcal{M}}}(\Lambda(o_t|z) > \tau) \leq \frac{\mathbb{E}_{o_t \sim P_{\text{in}}^{\mathcal{M}}} [e^{(\alpha-1)\Lambda(o_t|z)}]}{e^{(\alpha-1)\tau}} \quad (18)$$

Expanding the definition of $\Lambda(o_t|z)$ within the expectation, we have:

$$\mathbb{E}_{o_t \sim P_{\text{in}}^{\mathcal{M}}} \left[e^{(\alpha-1) \ln \left(\frac{P_{\text{in}}^{\mathcal{M}}(o_t)}{P_{\text{out}}^{\mathcal{M}}(o_t)} \right)} \right] = \mathbb{E}_{o_t \sim P_{\text{in}}^{\mathcal{M}}} \left[\left(\frac{P_{\text{in}}^{\mathcal{M}}(o_t)}{P_{\text{out}}^{\mathcal{M}}(o_t)} \right)^{\alpha-1} \right]$$

By the definition of Rényi divergence, $D_\alpha(P_{\text{in}}^{\mathcal{M}} \parallel P_{\text{out}}^{\mathcal{M}}) = \frac{1}{\alpha-1} \ln \mathbb{E}_{P_{\text{in}}^{\mathcal{M}}} \left[\left(\frac{P_{\text{in}}^{\mathcal{M}}(o_t)}{P_{\text{out}}^{\mathcal{M}}(o_t)} \right)^{\alpha-1} \right]$. Exponentiating both sides to isolate the expectation yields:

$$\mathbb{E}_{o_t \sim P_{\text{in}}^{\mathcal{M}}} \left[\left(\frac{P_{\text{in}}^{\mathcal{M}}(o_t)}{P_{\text{out}}^{\mathcal{M}}(o_t)} \right)^{\alpha-1} \right] = \exp \left((\alpha-1) D_\alpha(P_{\text{in}}^{\mathcal{M}} \parallel P_{\text{out}}^{\mathcal{M}}) \right)$$

Substituting this equivalence back into the numerator of equation 18 gives:

$$\Pr_{o_t \sim P_{\text{in}}^{\mathcal{M}}}(\Lambda(o_t|z) > \tau) \leq \frac{\exp \left((\alpha-1) D_\alpha(P_{\text{in}}^{\mathcal{M}} \parallel P_{\text{out}}^{\mathcal{M}}) \right)}{\exp \left((\alpha-1)\tau \right)}$$

Combining the exponents completes the proof:

$$\Pr_{o_t \sim P_{\text{in}}^{\mathcal{M}}}(\Lambda(o_t|z) > \tau) \leq \exp \left(-(\alpha-1)(\tau - D_\alpha(P_{\text{in}}^{\mathcal{M}} \parallel P_{\text{out}}^{\mathcal{M}})) \right)$$

□

D.2 RESULTS FOR GAUSSIAN MECHANISM

Lemma 11. *Let $\mathcal{M}$ be a Gaussian mechanism for mean estimation over a dataset Z_t of size N_t , which adds noise scaled to the population such that the variance is $\frac{\gamma^2}{N_t} I$. Let $P_{\text{in}}^{\mathcal{M}}(\cdot \mid Z_t)$ and $P_{\text{out}}^{\mathcal{M}}(\cdot \mid Z_t)$ denote the conditional distributions of the mechanism's output on adjacent datasets Z_t and $Z_t \setminus \{z\}$, respectively. For any order $\alpha > 1$, the Rényi divergence between these distributions is given by:*

$$D_\alpha(P_{\text{in}}^{\mathcal{M}}(\cdot \mid Z_t) \parallel P_{\text{out}}^{\mathcal{M}}(\cdot \mid Z_t)) = \frac{\alpha N_t}{2\gamma^2(N_t-1)^2} \|z - \mu_z\|_2^2$$

Proof. Let the output distributions of the mechanism on the adjacent datasets be defined as $P_{\text{in}}^{\mathcal{M}}(\cdot \mid Z_t) = \mathcal{N}(\mu_z, C_\gamma)$ and $P_{\text{out}}^{\mathcal{M}}(\cdot \mid Z_t) = \mathcal{N}(\mu_{-z}, C_\gamma)$, with shared covariance matrix $C_\gamma = \frac{\gamma^2}{N_t} I_{d \times d}$.

Assuming the mechanism computes the mean using the respective dataset sizes, the true, un-noised mean vectors and their difference are:

$$\mu_z = \frac{1}{N_t} \sum_{x \in Z_t} x, \quad \mu_{-z} = \frac{1}{N_t-1} \sum_{x \in Z_t \setminus \{z\}} x, \quad \mu_z - \mu_{-z} = \frac{z - \mu_z}{N_t-1}$$

For two multivariate Gaussian distributions sharing the same covariance matrix C_γ , the Rényi divergence of order α simplifies to the scaled Mahalanobis distance between their means. Substituting

the known difference and the inverse covariance matrix $C_\gamma^{-1} = \frac{N_t}{\gamma^2} I$, we obtain:

$$\begin{aligned}
 D_\alpha(P_{\text{in}}^{\mathcal{M}}(\cdot | Z_t) || P_{\text{out}}^{\mathcal{M}}(\cdot | Z_t)) &= \frac{\alpha}{2} (\mu_z - \mu_{-z})^\top C_\gamma^{-1} (\mu_z - \mu_{-z}) \\
 &= \frac{\alpha}{2} \left(\frac{z - \mu_z}{N_t - 1} \right)^\top \left(\frac{N_t}{\gamma^2} I \right) \left(\frac{z - \mu_z}{N_t - 1} \right) \\
 &= \frac{\alpha}{2} \cdot \frac{N_t}{\gamma^2} \cdot \frac{1}{(N_t - 1)^2} (z - \mu_z)^\top I (z - \mu_z) \\
 &= \frac{\alpha N_t}{2\gamma^2 (N_t - 1)^2} \|z - \mu_z\|_2^2
 \end{aligned}$$

□

Lemma 6. Let a mechanism output o_t be perturbed by additive isotropic Gaussian noise $\eta_t \sim \mathcal{N}(0, \frac{\gamma^2}{N_t} I_{d \times d})$, representing the noise. The log-likelihood ratio $\Lambda(o_t | z)$ conditioned on the dataset Z_t is given by:

$$\Lambda(o_t | z) | Z_t = \frac{N_t \|z - \mu_z\|^2}{2(N_t - 1)^2 \gamma^2} + \frac{N_t \langle z - \mu_z, \eta_t \rangle}{(N_t - 1) \gamma^2}$$

And it is distributed as:

$$\Lambda(o_t | z) | Z_t \sim \mathcal{N} \left(\frac{N_t \|z - \mu_z\|^2}{2(N_t - 1)^2 \gamma^2}, \frac{N_t \|z - \mu_z\|^2}{(N_t - 1)^2 \gamma^2} \right)$$

Proof. By definition, the log-likelihood ratio compares the probability density of observing the noisy output o_t when z is present (with mean μ_z) versus when z is absent (with mean μ_{-z}). Because the aggregated additive noise η_t is Gaussian with distribution $\mathcal{N}(0, \frac{\gamma^2}{N_t} I)$, the log-likelihood conditioned on Z_t evaluates to:

$$\begin{aligned}
 \Lambda(o_t | z) | Z_t &= \ln \left(\frac{\exp \left(-\frac{N_t}{2\gamma^2} \|o_t - \mu_z\|^2 \right)}{\exp \left(-\frac{N_t}{2\gamma^2} \|o_t - \mu_{-z}\|^2 \right)} \right) \\
 &= \frac{N_t}{2\gamma^2} \left(\|o_t - \mu_{-z}\|^2 - \|o_t - \mu_z\|^2 \right)
 \end{aligned}$$

Substituting the noisy observation $o_t = \mu_z + \eta_t$ and the shift $\mu_{-z} = \mu_z - \frac{z - \mu_z}{N_t - 1}$:

$$\begin{aligned}
 \Lambda(o_t | z) | Z_t &= \frac{N_t}{2\gamma^2} \left(\left\| \mu_z + \eta_t - \left(\mu_z - \frac{z - \mu_z}{N_t - 1} \right) \right\|^2 - \|\mu_z + \eta_t - \mu_z\|^2 \right) \\
 &= \frac{N_t}{2\gamma^2} \left(\left\| \frac{z - \mu_z}{N_t - 1} + \eta_t \right\|^2 - \|\eta_t\|^2 \right) \\
 &= \frac{N_t}{2\gamma^2} \left(\frac{\|z - \mu_z\|^2}{(N_t - 1)^2} + \frac{2}{N_t - 1} \langle z - \mu_z, \eta_t \rangle + \|\eta_t\|^2 - \|\eta_t\|^2 \right) \\
 &= \frac{N_t \|z - \mu_z\|^2}{2(N_t - 1)^2 \gamma^2} + \frac{N_t \langle z - \mu_z, \eta_t \rangle}{(N_t - 1) \gamma^2}
 \end{aligned}$$

Conditioned on Z_t , the term $\frac{N_t \|z - \mu_z\|^2}{2(N_t - 1)^2 \gamma^2}$ is a deterministic scalar constant. The sole source of randomness is the inner product with the aggregated noise vector $\eta_t \sim \mathcal{N}(0, \frac{\gamma^2}{N_t} I)$. This linear combination $\langle z - \mu_z, \eta_t \rangle$ forms a univariate Gaussian with mean 0 and variance

$$\frac{\gamma^2}{N_t} \|z - \mu_z\|^2.$$

Scaling this by $\frac{N_t}{(N_t - 1) \gamma^2}$, the conditional variance reduces to:

$$\mathbb{V} \left(\frac{N_t \langle z - \mu_z, \eta_t \rangle}{(N_t - 1) \gamma^2} \middle| Z_t \right) = \frac{N_t^2}{(N_t - 1)^2 \gamma^4} \left(\frac{\gamma^2}{N_t} \|z - \mu_z\|^2 \right) = \frac{N_t \|z - \mu_z\|^2}{(N_t - 1)^2 \gamma^2}$$

Thus,

$$\Lambda(o_t|z) \mid Z_t \sim \mathcal{N}\left(\frac{N_t\|z - \mu_z\|^2}{2(N_t - 1)^2\gamma^2}, \frac{N_t\|z - \mu_z\|^2}{(N_t - 1)^2\gamma^2}\right).$$

This completes the proof. $\square$

Lemma 7. *Given the aggregated empirical mean mechanism, the exact probability that a specific individual data point z is purged (i.e., its log-likelihood ratio breaches the threshold τ) conditioned on Z_t is given in closed form by:*

$$\Pr(\Lambda(o_t|z) > \tau \mid Z_t) = 1 - \Phi\left(\frac{\tau(N_t - 1)\gamma}{\sqrt{N_t}\|z - \mu_z\|} - \frac{\sqrt{N_t}\|z - \mu_z\|}{2(N_t - 1)\gamma}\right)$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

Proof. From lemma 6, the conditional log-likelihood ratio for a specific data point z and μ_z strictly follows a Gaussian distribution:

$$\Lambda(o_t|z) \mid Z_t \sim \mathcal{N}(\mu'_z, \sigma_z^2) \quad \text{where} \quad \mu'_z = \frac{N_t\|z - \mu_z\|^2}{2(N_t - 1)^2\gamma^2}, \quad \sigma_z = \frac{\sqrt{N_t}\|z - \mu_z\|}{(N_t - 1)\gamma} \quad (19)$$

The probability that $\Lambda(o_t|z)$ exceeds τ is precisely the tail probability of this Gaussian distribution. We evaluate this using the standard normal cumulative distribution function Φ :

$$\begin{aligned} \Pr(\Lambda(o_t|z) > \tau \mid Z_t) &= \Pr\left(\frac{\Lambda(o_t|z) - \mu'_z}{\sigma_z} > \frac{\tau - \mu'_z}{\sigma_z} \mid Z_t\right) \\ &= 1 - \Phi\left(\frac{\tau - \mu'_z}{\sigma_z}\right) \end{aligned}$$

Substituting the explicit formulae for μ'_z and σ_z from equation 19:

$$\begin{aligned} \Pr(\Lambda(o_t|z) > \tau \mid Z_t) &= 1 - \Phi\left(\frac{\tau - \frac{N_t\|z - \mu_z\|^2}{2(N_t - 1)^2\gamma^2}}{\frac{\sqrt{N_t}\|z - \mu_z\|}{(N_t - 1)\gamma}}\right) \\ &= 1 - \Phi\left(\frac{\tau(N_t - 1)\gamma}{\sqrt{N_t}\|z - \mu_z\|} - \frac{N_t\|z - \mu_z\|^2(N_t - 1)\gamma}{2(N_t - 1)^2\gamma^2\sqrt{N_t}\|z - \mu_z\|}\right) \\ &= 1 - \Phi\left(\frac{\tau(N_t - 1)\gamma}{\sqrt{N_t}\|z - \mu_z\|} - \frac{\sqrt{N_t}\|z - \mu_z\|}{2(N_t - 1)\gamma}\right) \end{aligned}$$

This completes the proof. $\square$

D.3 POPULATION EVOLUTION AND CRITICAL NOISE THRESHOLD

Lemma 8. *Given assumption 1 and the performative update rule in mechanism 1, the expected population size at round t , conditioned on the previous population N_{t-1} , is given by:*

$$\mathbb{E}[N_t \mid N_{t-1}] = N_{t-1}\left(1 + r - (q + r)p_\tau^{(t-1)}\right)$$

Proof. Let $\mathbb{1}(\Lambda(o_{t-1}|z_i) > \tau)$ be the indicator variable denoting whether the i -th data point experiences a privacy leakage exceeding the threshold τ at round $t - 1$. The exact performative update rule for the population size is defined as:

$$N_t = N_{t-1} - q \sum_{i=1}^{N_{t-1}} \mathbb{1}(\Lambda(o_{t-1}|z_i) > \tau) + r \left(N_{t-1} - \sum_{i=1}^{N_{t-1}} \mathbb{1}(\Lambda(o_{t-1}|z_i) > \tau)\right)$$

Where $z_i \in Z_{t-1}$. Grouping the terms yields:

$$N_t = N_{t-1}(1 + r) - (q + r) \sum_{i=1}^{N_{t-1}} \mathbb{1}(\Lambda(o_{t-1}|z_i) > \tau)$$

Taking the conditional expectation given N_{t-1} , we evaluate the joint expectation over the entire dataset $Z_{t-1} = \{z_1, \dots, z_{N_{t-1}}\}$ and the mechanism's noise η_{t-1} :

$$\mathbb{E}[N_t | N_{t-1}] = N_{t-1}(1+r) - (q+r)\mathbb{E}_{Z_{t-1}, \eta_{t-1}} \left[\sum_{i=1}^{N_{t-1}} \mathbb{1}(\Lambda(o_{t-1}|z_i) > \tau) \right]$$

By Fubini's theorem, we distribute the expectation inside the sum. As established, $\Lambda(o_{t-1}|z_i)$ for a specific point depends on z_i , μ_z , and the noise η_{t-1} . We evaluate the expectation by utilizing the tower property of expectation:

$$\mathbb{E}_{Z_{t-1}, \eta_{t-1}} \left[\sum_{i=1}^{N_{t-1}} \mathbb{1}(\Lambda(o_{t-1}|z_i) > \tau) \right] = \sum_{i=1}^{N_{t-1}} \mathbb{E}_{Z_{t-1}} [\Pr(\Lambda(o_{t-1}|z_i) > \tau | Z_{t-1})]$$

From lemma 7, the conditional probability of leakage for any specific data point z is given as:

$$\Pr(\Lambda(o_{t-1}|z) > \tau | Z_{t-1}) = 1 - \Phi \left(\frac{\tau(N_{t-1}-1)\gamma}{\sqrt{N_{t-1}}\|z - \mu_z\|} - \frac{\sqrt{N_{t-1}}\|z - \mu_z\|}{2(N_{t-1}-1)\gamma} \right)$$

Since all N_{t-1} data points are identically distributed, we proceed as follows:

$$\begin{aligned} \mathbb{E}[N_t | N_{t-1}] &= N_{t-1}(1+r) \\ &\quad - (q+r)N_{t-1}\mathbb{E}_{Z_{t-1}} \left[1 - \Phi \left(\frac{\tau(N_{t-1}-1)\gamma}{\sqrt{N_{t-1}}\|z - \mu_z\|} - \frac{\sqrt{N_{t-1}}\|z - \mu_z\|}{2(N_{t-1}-1)\gamma} \right) \right] \end{aligned}$$

Since mechanism $\mathcal{M}$ in algorithm 1 clips the samples to $\bar{z}$, we denote the empirical mean on the clipped samples as $\mu_{\bar{z}}$. Thus, the marginal leakage probability for z is

$$p_{\tau}^{(t-1)} = \mathbb{E}_{Z_{t-1}} \left[1 - \Phi \left(\frac{\tau(N_{t-1}-1)\gamma}{\sqrt{N_{t-1}}\|\bar{z} - \mu_{\bar{z}}\|} - \frac{\sqrt{N_{t-1}}\|\bar{z} - \mu_{\bar{z}}\|}{2(N_{t-1}-1)\gamma} \right) \right].$$

Which expands as shown in equation 11. Therefore, our final expression is:

$$\begin{aligned} \mathbb{E}[N_t | N_{t-1}] &= N_{t-1}(1+r) - (q+r)N_{t-1}p_{\tau}^{(t-1)} \\ &= N_{t-1} \left(1+r - (q+r)p_{\tau}^{(t-1)} \right) \end{aligned}$$

This completes the proof. $\square$

Theorem 2. *To prevent demographic collapse and guarantee the expected population survives and continues to grow (maintaining a growth rate $C(\gamma) \geq 1$), the mechanism's noise scale γ must strictly satisfy the explicit lower bound $\gamma \geq \gamma_c$. Given an ℓ_2 clipping bound $R = \Omega(\sqrt{d})$, the critical threshold γ_c is defined as:*

$$\gamma_c \triangleq \Omega \left(\frac{\sqrt{dN_{t-1}}}{N_{t-1}-1} \left(\frac{\Phi_{q,r}^{-1} + \sqrt{(\Phi_{q,r}^{-1})^2 + 2\tau}}{2\tau} \right) \right)$$

where $\Phi_{q,r}^{-1} \triangleq \Phi^{-1} \left(\frac{q}{q+r} \right)$ is the Probit function evaluated at the performative survival threshold.

Proof. To ensure the user population survives without decaying, the expected growth rate must be at least 1: $C(\gamma) \geq 1$. Given the definition of the performative growth factor, this restricts the maximum permissible expected marginal leakage probability p_{τ} :

$$C(\gamma) \geq 1 \implies 1+r - (q+r)p_{\tau} \geq 1 \implies p_{\tau} \leq \frac{r}{q+r}$$

Worst-case leakage occurs at the ℓ_2 clipping bound R . Thus,

$$\|\bar{z} - \mu_{\bar{z}}\| \leq 2R.$$

Hence, the worst-case leakage satisfies:

$$1 - \Phi \left(\frac{\tau(N_{t-1} - 1)\gamma}{2R\sqrt{N_{t-1}}} - \frac{R\sqrt{N_{t-1}}}{(N_{t-1} - 1)\gamma} \right) \leq \frac{r}{q + r}.$$

Rearranging to isolate the CDF and applying the Probit function (Φ^{-1}) yields:

$$\frac{\tau(N_{t-1} - 1)\gamma}{2R\sqrt{N_{t-1}}} - \frac{R\sqrt{N_{t-1}}}{(N_{t-1} - 1)\gamma} \geq \Phi^{-1} \left(1 - \frac{r}{q + r} \right) = \Phi^{-1} \left(\frac{q}{q + r} \right) \triangleq \Phi_{q,r}^{-1}. \quad (20)$$

Let

$$u = \frac{(N_{t-1} - 1)\gamma}{2R\sqrt{N_{t-1}}}.$$

Substituting u transforms equation 20 into a standard quadratic equation:

$$\tau u - \frac{1}{2u} \geq \Phi_{q,r}^{-1} \implies 2\tau u^2 - 2\Phi_{q,r}^{-1}u - 1 \geq 0.$$

Taking the valid positive root of the quadratic provides the strict lower bound for u :

$$u \geq \frac{2\Phi_{q,r}^{-1} + \sqrt{4(\Phi_{q,r}^{-1})^2 + 8\tau}}{4\tau} = \frac{\Phi_{q,r}^{-1} + \sqrt{(\Phi_{q,r}^{-1})^2 + 2\tau}}{2\tau}. \quad (21)$$

Substituting $u = \frac{(N_{t-1} - 1)\gamma}{2R\sqrt{N_{t-1}}}$ back into equation 21 and using $R = \Omega(\sqrt{d})$, we get:

$$\begin{aligned} \gamma &\geq \frac{2R\sqrt{N_{t-1}}}{N_{t-1} - 1} \left(\frac{\Phi_{q,r}^{-1} + \sqrt{(\Phi_{q,r}^{-1})^2 + 2\tau}}{2\tau} \right) \triangleq \gamma_c \\ &= \Omega \left(\frac{\sqrt{d}N_{t-1}}{N_{t-1} - 1} \left(\frac{\Phi_{q,r}^{-1} + \sqrt{(\Phi_{q,r}^{-1})^2 + 2\tau}}{2\tau} \right) \right) \end{aligned}$$

This establishes the minimum noise necessary to sustain the population explicitly in terms of the dimensionality d , completing the proof. $\square$

D.4 BOUNDS AND ASYMPTOTICS OF THE ERROR

Theorem 3. Assume that the population is floored at one participant ($N_t \geq 1$). Conditioned on N_{t-1} , the expected squared Mahalanobis error of the private empirical mean o_t is bounded from below and above by:

$$\frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{C_t(\gamma)N_{t-1}} \leq \mathbb{E} [\|o_t - \mu\|_{\Sigma^{-1}}^2 \mid N_{t-1}] \leq \kappa_{q,r} \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{C_t(\gamma)N_{t-1}} \quad (22)$$

$$\text{with } \kappa_{q,r} \triangleq \frac{(2 + r - q)^2}{4(1 - q)(1 + r)} = \mathcal{O} \left(\frac{1}{1 - q} \right), \text{ since } 0 \leq r, q \leq 1$$

where the asymptotic notation in the upper bound is particularly useful in the regime $q \rightarrow 1$.

Proof. Recall that the exact squared Mahalanobis error, conditioned strictly on a known population size N_t , isolates the population in the denominator:

$$\mathbb{E} [\|o_t - \mu\|_{\Sigma^{-1}}^2 \mid N_t] = \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_t}.$$

Taking the conditional expectation given N_{t-1} gives

$$\mathbb{E} [\|o_t - \mu\|_{\Sigma^{-1}}^2 \mid N_{t-1}] = \left(\gamma^2 \text{Tr}(\Sigma^{-1}) + d \right) \mathbb{E} \left[\frac{1}{N_t} \mid N_{t-1} \right].$$

Lower Bound. We apply Jensen's inequality to the convex function $x \mapsto 1/x$ on $(0, \infty)$:

$$\mathbb{E} \left[\frac{1}{N_t} \middle| N_{t-1} \right] \geq \frac{1}{\mathbb{E}[N_t | N_{t-1}]} = \frac{1}{C(\gamma)N_{t-1}}.$$

Upper Bound. From the performative update rule, the ratio N_t/N_{t-1} is a random variable taking values strictly in the interval $[1 - q, 1 + r]$. By the Kantorovich inequality for the convex function $x \mapsto 1/x$, the expectation of its inverse is bounded by:

$$\mathbb{E} \left[\frac{N_{t-1}}{N_t} \middle| N_{t-1} \right] \leq \frac{((1 - q) + (1 + r))^2}{4(1 - q)(1 + r)} \frac{1}{\mathbb{E} \left[\frac{N_t}{N_{t-1}} \middle| N_{t-1} \right]}.$$

Substituting $\mathbb{E}[N_t | N_{t-1}] = C(\gamma)N_{t-1}$ and dividing by N_{t-1} directly yields:

$$\mathbb{E} \left[\frac{1}{N_t} \middle| N_{t-1} \right] \leq \frac{(2 + r - q)^2}{4(1 - q)(1 + r)} \frac{1}{C(\gamma)N_{t-1}} = \frac{\kappa_{q,r}}{C(\gamma)N_{t-1}}.$$

Combined Error Bounds. Multiplying the lower and upper bounds on the expected inverse population by $(\gamma^2 \text{Tr}(\Sigma^{-1}) + d)$ establishes equation 22. $\square$

Corollary 1 (Zero-Noise Collapse). *For any strictly positive purge rate $q > 0$, as the algorithmic noise vanishes ($\gamma \rightarrow 0^+$), the expected estimation error diverges to infinity over time under the mean field approximation of N_t .*

Proof. As $\gamma \rightarrow 0^+$, the log-likelihood ratio diverges for any z with $\|z - \mu_z\| > 0$, driving the marginal leakage probability $p_\tau^{(t)} \rightarrow 1$ for all t . The demographic growth factor reduces to $C_t(\gamma) \rightarrow 1 - q$ for all t , resulting in an expected population of $N_0(1 - q)^t$. By the lower bound in equation 22, the expected estimation error is at least of order $\frac{d}{N_0(1 - q)^t}$. For $q > 0$, the denominator decays to zero, causing the error to diverge as t increases. $\square$

Lemma 12. *Under mean-field approximation, if $C_t(\gamma) > 1$ for all t (i.e., the expected population strictly grows), then $C_t(\gamma)$ is a strictly increasing sequence.*

Proof. Under the mean-field approximation of N_t , the expected population size at step t is given by $N_t = C_t(\gamma)N_{t-1}$. Since we are given that $C_t(\gamma) > 1$ for all t , the population strictly grows, meaning $N_t > N_{t-1}$ (and consequently its inverse shrinks: $1/N_t < 1/N_{t-1}$).

As N_t grows and $1/N_t$ shrinks, by the definition of $p_\tau^{(t-1)}$, it strictly increases. Thus:

$$p_\tau^{(t-1)} > p_\tau^{(t)}$$

To determine the behavior of $C_t(\gamma)$, we evaluate the difference between consecutive terms:

$$C_{t+1}(\gamma) - C_t(\gamma) = \left[1 + r - (q + r)p_\tau^{(t)} \right] - \left[1 + r - (q + r)p_\tau^{(t-1)} \right]$$

Canceling the $1 + r$ terms and factoring out $(q + r)$ yields:

$$C_{t+1}(\gamma) - C_t(\gamma) = (q + r) \left(p_\tau^{(t-1)} - p_\tau^{(t)} \right)$$

Since $q + r > 0$ and we established that $p_\tau^{(t-1)} - p_\tau^{(t)} > 0$, the product of these two positive terms must be positive:

$$C_{t+1}(\gamma) - C_t(\gamma) > 0$$

Therefore, $C_t(\gamma)$ is a strictly increasing sequence. $\square$

Corollary 2 (Infinite Horizon Asymptotics). *For any positive recruitment rate $r > 0$ and any fixed noise scale γ such that $C_t(\gamma) > 1$ for all t , the expected estimation error at round t converges to zero as $t \rightarrow \infty$, at the geometric rate $\mathcal{O}(C_1(\gamma)^{-t})$ under the mean field approximation of N_t . This rate is fastest in the infinite-noise limit ($\gamma \rightarrow \infty$), where $C(\gamma) \rightarrow 1 + r$.*

Proof. For a fixed γ with $C_t(\gamma) > 1$ for all t , iterating the bounds in equation 22 (under mean-field approximation of N_t), $\mathcal{E}_t(\gamma)$ is at most $\kappa_{q,r} \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0 \prod_{k=1}^t C_k(\gamma)}$. Due to lemma 12, this quantity is upper bounded by $\kappa_{q,r} \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0 C_1(\gamma)^t}$ and decays to zero as $t \rightarrow \infty$. As $\gamma \rightarrow \infty$, the marginal leakage probability vanishes ($p_r \rightarrow 0$), maximizing the growth factor to $C(\gamma) \rightarrow 1 + r$ and the expected population to $N_0(1 + r)^t$: the exponential expansion of the population eventually dominates the fixed-in- t mechanism variance term γ^2 .

This conclusion continues to hold when q is large (even as $q \rightarrow 1$) and $r > 0$ is small but nonzero, provided $C(\gamma) > 1$. Although $\kappa_{q,r} = \mathcal{O}((1 - q)^{-1})$ may become large as $q \rightarrow 1$, it remains independent of t , while $C_1(\gamma)^t$ grows exponentially. Thus, the exponential term eventually dominates the polynomial numerator $\gamma^2 \text{Tr}(\Sigma^{-1}) + d$ and the multiplicative factor $\kappa_{q,r}$, so the expected estimation error still converges to zero as $t \rightarrow \infty$. $\square$

Lemma 9. For a finite operational timeline T and an initial population N_0 , let $C_t(\gamma) > 1$ denote the strictly positive expected per-round demographic growth factor for all t . The cumulative expected Mahalanobis error aggregated across all T rounds, defined as $\mathcal{E}_{\text{total}}(\gamma) \triangleq \sum_{t=1}^T \mathcal{E}_t(\gamma)$ is upper bounded by $\mathcal{E}_{\text{total}}(\gamma) \leq \hat{\mathcal{E}}_{\text{total}}(\gamma)$, where:

$$\hat{\mathcal{E}}_{\text{total}}(\gamma) \triangleq \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0(C_1(\gamma) - 1)} \left(1 - \frac{1}{[C_1(\gamma)]^T} \right) \quad (23)$$

Proof. To evaluate the true cost of privacy over a finite horizon T , we must aggregate the expected error across all rounds. The cumulative expected Mahalanobis error is the sum of the single-round errors:

$$\mathcal{E}_{\text{total}}(\gamma) = \sum_{t=1}^T \mathbb{E} [\|o_t - \mu\|_{\Sigma^{-1}}^2] = \sum_{t=1}^T \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0 \prod_{k=1}^t C_k(\gamma)} \stackrel{(a)}{\leq} \sum_{t=1}^T \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0 [C_1(\gamma)]^t}$$

Where (a) holds due to lemma 12. We apply the formula for sum of a finite geometric series:

$$\begin{aligned} \hat{\mathcal{E}}_{\text{total}}(\gamma) &\triangleq \sum_{t=1}^T \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0 [C_1(\gamma)]^t} \\ &= \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0} \left(\frac{1 - \frac{1}{[C_1(\gamma)]^T}}{1 - \frac{1}{C_1(\gamma)}} \right) \frac{1}{C_1(\gamma)} \\ &= \frac{\gamma^2 \text{Tr}(\Sigma^{-1}) + d}{N_0(C_1(\gamma) - 1)} \left(1 - \frac{1}{[C_1(\gamma)]^T} \right) \end{aligned}$$

This completes the proof. $\square$

E PAC GUARANTEE FOR MAHALANOBIS ERROR

To quantify the accuracy of the noisy empirical mean, we use Mahalanobis error, a scale-invariant measure that accounts for covariance structure. We derive a PAC guarantee giving an explicit finite-sample, high-probability bound on the combined effects of sampling variability and Gaussian noise, providing a baseline for general covariance structures.

Theorem 4. Assume $\Sigma \succ 0$, $C_1(\gamma) > 1$ which by theorem 2 is true when $\gamma \geq \gamma_c$, and also assume by mean field approximation, $N_t \geq N_0(C_1(\gamma))^t$. Define

$$\begin{aligned} M &\triangleq \sqrt{\|\mu\|_2^2 + \text{Tr}(\Sigma)}, \quad \mathcal{B}(R) \triangleq \sqrt{\lambda_{\max}(\Sigma^{-1})} M \exp \left(-\frac{(R - M)^2}{4\lambda_{\max}(\Sigma)} \right), \\ \mathcal{V}_{\delta/2} &\triangleq 2R\sqrt{2\lambda_{\max}(\Sigma^{-1}) \ln \frac{4}{\delta}}, \quad \mathcal{K}_{\delta/2} \triangleq d + 2\sqrt{d \ln \frac{2}{\delta}} + 2 \ln \frac{2}{\delta}. \end{aligned}$$

For any $\delta \in (0, 1)$ and $\epsilon > \mathcal{B}(R)$, if

$$t \geq \frac{1}{\ln C_1(\gamma)} \ln \left(\frac{(\mathcal{V}_{\delta/2} + \gamma\sqrt{\lambda_{\max}(\Sigma^{-1})\mathcal{K}_{\delta/2}})^2}{N_0(\epsilon - \mathcal{B}(R))^2} \right), \quad (24)$$

then $D_M \triangleq \|o_t - \mu\|_{\Sigma^{-1}} \leq \epsilon$ with probability at least $1 - \delta$.

Proof. We know, $\mu_{\bar{z}} \triangleq \frac{1}{N_t} \sum_{i=1}^{N_t} \bar{z}_i$.

By the triangle inequality,

$$D_M \leq \|\mathbb{E}[\bar{z}_i] - \mu\|_{\Sigma^{-1}} + \|\mu_{\bar{z}} - \mathbb{E}[\bar{z}_i]\|_{\Sigma^{-1}} + \|\eta_t\|_{\Sigma^{-1}}. \quad (25)$$

Bounding the Clipping bias. Using $\mu = \mathbb{E}[z_i]$,

$$\begin{aligned} \|\mathbb{E}[\bar{z}_i] - \mu\|_{\Sigma^{-1}} &\leq \sqrt{\lambda_{\max}(\Sigma^{-1})} \mathbb{E}\|\bar{z}_i - z_i\|_2 \\ &\leq \sqrt{\lambda_{\max}(\Sigma^{-1})} \sqrt{\mathbb{E}\|z_i\|_2^2} \sqrt{\Pr(\|z_i\|_2 > R)} \\ &= \sqrt{\lambda_{\max}(\Sigma^{-1})} M \sqrt{\Pr(\|z_i\|_2 > R)}. \end{aligned}$$

Since $z \mapsto \|z\|_2$ is 1-Lipschitz,

$$\Pr(\|z_i\|_2 > R) \leq \exp\left(-\frac{(R - M)^2}{2\lambda_{\max}(\Sigma)}\right),$$

for $R > M$. Hence

$$\|\mathbb{E}[\bar{z}_i] - \mu\|_{\Sigma^{-1}} \leq \mathcal{B}(R). \quad (26)$$

Bounding the Sample variation. For $Y_i = \bar{z}_i - \mathbb{E}[\bar{z}_i]$, we have

$$\mathbb{E}[Y_i] = 0, \quad \|Y_i\|_{\Sigma^{-1}} \leq 2R\sqrt{\lambda_{\max}(\Sigma^{-1})}.$$

Pinelis' inequality therefore gives, with probability at least $1 - \delta/2$,

$$\|\mu_{\bar{z}} - \mathbb{E}[\bar{z}_i]\|_{\Sigma^{-1}} \leq \frac{\mathcal{V}_{\delta/2}}{\sqrt{N_t}}. \quad (27)$$

Bounding the Injected noise. Writing $\eta_t = \gamma\zeta/\sqrt{N_t}$ with $\zeta \sim \mathcal{N}(0, I_d)$,

$$\|\eta_t\|_{\Sigma^{-1}}^2 \leq \frac{\gamma^2 \lambda_{\max}(\Sigma^{-1})}{N_t} \|\zeta\|_2^2.$$

By the Laurent–Massart inequality, with probability at least $1 - \delta/2$,

$$\|\eta_t\|_{\Sigma^{-1}} \leq \frac{\gamma \sqrt{\lambda_{\max}(\Sigma^{-1}) \mathcal{K}_{\delta/2}}}{\sqrt{N_t}}. \quad (28)$$

Combine. By the union bound, equation 27 and equation 28 hold simultaneously with probability at least $1 - \delta$. Combining them with equation 25 and equation 26,

$$\begin{aligned} D_M &\leq \mathcal{B}(R) + \frac{\mathcal{V}_{\delta/2} + \gamma \sqrt{\lambda_{\max}(\Sigma^{-1}) \mathcal{K}_{\delta/2}}}{\sqrt{N_t}} \\ &\leq \mathcal{B}(R) + \frac{\mathcal{V}_{\delta/2} + \gamma \sqrt{\lambda_{\max}(\Sigma^{-1}) \mathcal{K}_{\delta/2}}}{\sqrt{N_0(C_1(\gamma))^t}}. \end{aligned}$$

Thus $D_M \leq \epsilon$ whenever

$$(C_1(\gamma))^t \geq \frac{(\mathcal{V}_{\delta/2} + \gamma \sqrt{\lambda_{\max}(\Sigma^{-1}) \mathcal{K}_{\delta/2}})^2}{N_0 (\epsilon - \mathcal{B}(R))^2}.$$

Since $C_1(\gamma) > 1$, taking logarithms gives the stated condition on t as given in equation 24. $\square$